Examination before attraction: lifelong affinities and detection-adjusted item appeal for recommendation under randomised exposure

Anonymous

Abstract

Recommender logs reflect the policy that produced them, but the target is a viewer's response to an item shown at random. We study two log-only signals on KuaiRand-27K, where videos are inserted uniformly at random. LEAF gives each user lifelong, exposure-conditional affinities for content keys. DAX estimates item attractiveness by EM on a click model that discounts non-clicks by user and scenario examination. Both are added to LightGCN, and DAX is also applied post hoc to other backbones. Under a temporal protocol with five user-level splits, LEAF-LGN with DAX reaches 0.4854 nDCG@10, against 0.4634 for the QP family and 0.4733 for the QP family with DAX. The paired per-user differences are +22.32 and +12.04 ($\times 10^{-3}$), with Holm-adjusted t test p values of 2.18e-124 and 7.92e-47, and Holm-adjusted Nadeau-Bengio p values of 0.0456 and 0.0223 over the splits. It also exceeds all other comparators with results, including DIN with DAX. Shuffling attractiveness across items removes the DAX gain, while shuffling within tags keeps 41.6%. The gain holds on KuaiRand-1K and across three training seeds. We do not claim unbiasedness, causal identification of item quality, generality beyond one platform family, or superiority over SIM with DAX, which is not yet run.

Keywords: recommender systems, exposure bias, click models, lifelong user behaviour, randomised exposure, graph collaborative filtering

1. Introduction

Recommenders are usually trained and evaluated on logs written by a production policy. Such a log records which items the policy chose to show to which users, and whether those users clicked. A model fitted to this

record learns the response of the audiences that the policy selected, and a model that ranks well on a held-out part of the same record is rewarded partly for reproducing the policy. The question that matters for an item is different: how a user who was not selected for the item would respond to it. Randomised exposure measures this target directly. In the KuaiRand logs (Gao et al., 2022b), a feed slot is replaced with small probability by a video drawn uniformly from a pool of 7,583 videos, so the inserted candidates reach users whatever the policy would have predicted.

We first ask where the remaining accuracy of a strong graph recommender lies on this target. The headroom is at the item level. A leave-one-user-out item oracle, which scores each candidate by the uniform click rate of the item among the other test users, reaches nDCG@10 of 0.5068 against 0.4735 for LEAF-LGN, a gap of $+33.31 \times 10^{-3}$. The oracle reads test labels, so it is a diagnostic and not a method. The gap is concentrated in the items with the highest uniform click rates. The post-cut log predicts the uniform rate no better than the pre-cut log (Spearman 0.405 against 0.431), so temporal drift is not the bottleneck. Personalisation fitted on uniform in-pool data adds nothing once item means are known. What the policy log lacks is a measurement of how appealing an item is to an audience the policy did not select.

We address this with two signals that are estimated from the log alone and added to a graph backbone. The first is a lifelong, exposure-conditional affinity (LEAF). For each user, LEAF uses the impressions of videos outside the test pool before the cut-off, and summarises, for every content key of a pool video (tag, author, music), how often the user clicked after an impression. Because the statistics are conditioned on the impression, they measure the response of the user given exposure, not how often the policy chose the key. The second is detection-adjusted attractiveness (DAX). DAX models each logged impression as an examination event, whose probability depends on the user and the scenario, followed by a click that depends on the attractiveness of the item. A non-click is then weak evidence against an item when the user rarely examines, and strong evidence when examination is likely. The attractiveness is fitted by expectation-maximisation on the pre-cut log and shrunk towards the mean of its content tag.

We make four contributions. Each is stated with the number that supports it. Section 5 gives the full tables, and each number is traced to a result file in the replication materials.

C1 (headroom diagnosis). On randomised insertions, the item oracle exceeds LEAF-LGN by $+33.31 \times 10^{-3}$ nDCG@10. This is a diagnostic that uses test labels and is never a selection input.

C2 (LEAF). Lifelong exposure-conditional channels built from out-of-pool videos transfer through content keys to a disjoint, randomly exposed pool. Added to LightGCN they give +15.46 [1.53] $\times 10^{-3}$ nDCG@10, and a control that shuffles the users' lifelong tables removes the lifelong gain. LEAF alone does not pass the pre-declared Stage 1 gate against the QP family (Section 5).

C3 (DAX). DAX is complementary to LEAF. On top of LEAF-LGN it gives +11.96 [1.47] $\times 10^{-3}$ nDCG@10, and the same post-hoc module is applied to EASE and to the QP family. The DAX weight is selected on validation together with the backbone epoch and the DAX configuration; it is not learned jointly.

C4 (main result). Under a temporal protocol with five user-level splits, LEAF-LGN+DAX reaches nDCG@10 of 0.4854 against 0.4634 for the QP family, a paired difference of +22.32 [1.83] $\times 10^{-3}$ with a positive difference in every split. The QP family is an IPS- and DR-weighted BPR LightGCN baseline that is described in full in Appendix A.

The comparison in C4 is not budget-matched. LEAF-LGN+DAX searches 3,840 configurations per split, whereas the QP family alone searches 20. The fair comparator QP family+DAX adds the same DAX module to the QP family and searches 3,200 configurations. LEAF-LGN+DAX still exceeds it, by +12.04 [1.63] $\times 10^{-3}$ nDCG@10, and we treat this difference as the closer comparison.

We do not claim that DAX recovers an unbiased or causally identified item quality, since its factors are identified only up to a scale; that LEAF-LGN+DAX outperforms SIM
citeppi2020sim or TWIN
citepchang2023twin, which were not run on the main setting; that the weight of DAX is learned end to end; or that the results hold beyond one platform family and one training seed per split. The deep lifelong baseline that we did run on the main setting is DIN (Zhou et al., 2018), which attends over the last L events of each user's lifelong sequence. LEAF-LGN+DAX exceeds DIN+DAX by +5.73 [1.64] $\times 10^{-3}$ nDCG@10 on the same splits.

Section 2 places the work among lifelong behaviour models, click models, exposure-aware recommenders, occupancy and small-area estimators, and debiased graph recommenders. Section 3 defines the LEAF channels, DAX and their combination. Section 4 gives the temporal protocol, the splits, the controls and the tests. Section 5 reports the headroom diagnosis, the main comparison, the backbone and ablation analyses and the robustness checks. Section 6 discusses why the multiplicative combination and the lifelong channels help. Section 7 lists the limitations, and Section 8 concludes.

2. Related work

We organise the related work by the question that each line of work answers. Each subsection ends with the difference from the present work. Table 1 collects the closest works, and the paragraph after it states the novelty claim.

2.1. Lifelong behaviour models

Lifelong behaviour models in click-through prediction retrieve information from long user histories. SIM (Pi et al., 2020) selects behaviours from a lifelong sequence that match the category of the target item, and passes them to a deep CTR model. TWIN (Chang et al., 2023) applies a two-stage target attention over a lifelong sequence on the Kuaishou platform, which belongs to the same platform family as KuaiRand. Both learn from policy logs and rely on item identifiers that the history shares with the target. *Difference.* LEAF uses exposure-conditional statistics in which non-clicks enter the denominator, transfers to a pool that is disjoint from the lifelong history through content keys only, and is evaluated on randomised insertions under a temporal protocol.

2.2. Click models and examination

A click model explains a click as the product of an examination event and a judgement about the item. The examination hypothesis and its position-based form (Richardson et al., 2007) make the examination probability depend on the position of the item. Craswell et al. (2008) compare cascade and position-bias models on search logs, and the monograph of Chuklin et al. (2015) summarises this family. The dynamic Bayesian network model (Chapelle and Zhang, 2009) separates perceived relevance from actual relevance through a satisfaction state. *Difference.* DAX conditions examination

on the user, through a lifelong prior, and on the scenario, through a tab factor; it has no position term and no satisfaction state, and it targets item appeal under uniform insertion, which we check on randomised exposure.

2.3. Exposure-aware recommendation

A line of work learns from impressions and not only from clicks. Impression-aware recommenders treat shown but unclicked items as evidence and use the impression set as features or as negative samples (Maurera et al., 2025). ExpoMF (Liang et al., 2016) models exposure as a latent variable that is fitted by EM. Inverse-propensity estimators for missing-not-at-random implicit feedback (Saito et al., 2020) weight the loss by estimated exposure propensities, and Joachims et al. (2017) apply the same idea to clicks with examination propensities. Wang et al. (2018) and Wang et al. (2016) estimate position bias and relevance jointly, without randomisation, in personal search. ESMM (Ma et al., 2018) models the entire impression space, and Krause et al. (2024) use multinomial-logit choice with item co-exposure. cEASE (Jeunen et al., 2020) adds item side information to a closed-form model. *Difference.* The impression channel alone is not claimed as new here, and DAX differs in that it is a generative model of examination and appeal fitted by EM, whereas the estimators above weight a loss or use impressions as negatives or features.

2.4. Occupancy, detection and small-area estimation

Occupancy models separate the presence of a species from its detection in repeated surveys (MacKenzie et al., 2002): a site can be occupied and still go undetected, and the pattern of repeated non-detections identifies both quantities. Small-area estimation shrinks noisy direct estimates for many small areas towards a regression on area covariates (Fay and Herriot, 1979), and multilevel regression with poststratification estimates population quantities from non-representative samples (Park et al., 2004). We found no recommendation paper that uses the occupancy likelihood to estimate item appeal from a policy log. *Difference.* The likelihood of DAX has the same structure as the occupancy model, with an impression playing the role of a survey and a click the role of a detection, but its estimand is item appeal under a uniform audience, and the borrowed MRP and Fay-Herriot forms, which we tested on this target, are weaker than the multiplicative form.

2.5. Debiasing for graph recommenders

Graph collaborative filtering propagates signals over the user-item graph, and bias corrections have been applied to that propagation. NAVIP (Kim et al., 2022) debiases neighbour aggregation in a graph neural network for recommendation. Popularity-bias work (Zhang et al., 2021) applies a causal intervention to the popularity effect. AutoDebias (Chen et al., 2021) learns debiasing parameters from a small uniform sample, which is not admissible under our protocol because the randomly exposed data serve only for validation and test. Two further graph methods, adversarial edge dropout and the removal of a popularity direction from the embeddings after training, are relevant here (Zhang et al., 2024; Islam et al., 2026); neither is re-run in this paper. The backbones used in our experiments are LightGCN (He et al., 2020), EASE (Steck, 2019) and the r-AdjNorm normalisation (Zhao et al., 2022). *Difference.* The graph methods above act on propagation or on the embeddings and use no impressions and no lifelong data, whereas DAX re-estimates item appeal from the impressions under a different audience, and LEAF adds user-side behaviour on other items.

2.6. Evaluation under randomised exposure

Random insertion is a direct test of how a recommender responds to items that the policy did not choose. KuaiRand (Gao et al., 2022b) ships randomly exposed videos in its feed logs, and KuaiRec (Gao et al., 2022a) is a fully observed matrix that supports the evaluation of recommenders. Several recent papers use the random insertions of KuaiRand for evaluation: propensity calibration (Hu et al., 2025), debiasing of watch-time signals (Zhao et al., 2023), an analysis of whether the evaluation of debiasing methods is itself biased (Wang et al., 2025), and an audit of category means in logs with random insertions (Zhang et al., 2026). Classical debiasing benchmarks use a small randomly exposed sample from Coat and Yahoo! R3 (Schnabel et al., 2016; Marlin and Zemel, 2009). *Difference.* As far as our searches went, none of these works uses the out-of-pool history of each user in the random test, which is the source of the LEAF channels here.

2.7. Positioning and novelty

The novelty claim is therefore a combination and its controlled evidence, not any single ingredient. Each ingredient has a precedent: lifelong behaviour statistics, examination multiplied by attractiveness, and Beta shrinkage towards a group mean. As far as our searches went, none of the works in

Table 1 evaluates the combination in the form used here, which has lifelong channels built from other pools, examination factors for the user and the scenario, and a temporal protocol on randomly inserted items, with tuned baselines and split-level inference.

3. Method

The method has two components, and both are computed from the log alone. LEAF (lifelong exposure-conditional affinities) gives each user a set of channel values on the content keys of a pool video. The values are computed from the user's impressions of other videos before the cut-off. LEAF-LGN adds these channels to a LightGCN backbone. DAX (detection-adjusted attractiveness) estimates an appeal for each pool item by fitting a two-factor Bernoulli model to the pre-cut standard log. The two components are combined with a weight that is selected on validation. The random-exposure log is never used to fit either component; it is used only to validate and test the scores (Section 4).

3.1. Setting, notation and leakage boundary

Let $\mathcal{U}$ be the set of users ($m = |\mathcal{U}|$), $\mathcal{I}$ the pool of videos that the random insertion can place ($n = |\mathcal{I}|$), and $\mathcal{T}$ the set of scenarios (tabs). Tab 1 is the scenario of the random test. The cut-off is T_0, 2022-04-22 00:00 UTC+8, which is $1\,650\,556\,800\,000$ ms. Every quantity that is used for training or for selection is computed from events with time strictly before T_0.

Pre-cut standard log. For a pair (u, i) let $O_{ui} = 1$ if u was shown i before T_0, and let $Y_{ui} = 1$ if at least one of these impressions was clicked. The clicked pairs are $X_{ui} = O_{ui}Y_{ui}$ and the pairs shown but not clicked are $N_{ui} = O_{ui}(1 - Y_{ui})$. Each impression r of the pre-cut log carries a user u_r, an item i_r, a tab t_r and a click $y_r \in \{0, 1\}$.

Lifelong log. For each user, the lifelong log consists of the impressions before T_0 on videos outside $\mathcal{I}$. Three content-key families are used, $\mathcal{K} = \{\text{tag}, \text{author}, \text{music}\}$. For a family k the map $v_k : \mathcal{I} \to \mathcal{V}_k \cup \{\varnothing\}$ gives the key value of a pool item: the first tag, the author identifier, or the music identifier. A missing value is mapped to $\varnothing$.

Test target. The random-exposure impressions of a user after T_0 form that user's candidate list. A candidate is a pool item that the user saw in the random log, and the label is the click. A pair that was shown before T_0 is never a candidate.

Boundary. The latest pre-cut training timestamp is 1 650 556 797 718 ms, which is before T_0. The earliest test timestamp is 1 650 595 332 831 ms, which is after T_0 (`results/logs_v11/dax_lgn_post.log`, line "temporal protocol"). The runners of the DAX and LEAF-LGN experiments assert this ordering (`drup/leaf.py check_temporal_protocol`). Each lifelong table and the DAX fit read only rows with $t < T_0$.

3.2. LEAF: lifelong exposure-conditional channels

For a user u, a key family k and a key value $v \in \mathcal{V}_k$, let n_{uv}^k and c_{uv}^k be the number of impressions and the number of clicks of u before T_0 on out-of-pool videos whose key value in family k is v. Let M_u and C_u be the number of impressions and clicks of u before T_0 on all out-of-pool videos. The lifelong base rate is

$$r_u = \frac{C_u + 1}{M_u + 2}. \tag{1}$$

Two statistics are formed for each user and key value. The first is the exposure-conditional lift, which shrinks the click rate of the key towards the base rate with strength $s = 5$:

$$\ell_{uv}^k = \frac{c_{uv}^k + s\, r_u}{n_{uv}^k + s} - r_u. \tag{2}$$

The second is the familiarity

$$f_{uv}^k = \log(1 + n_{uv}^k). \tag{3}$$

The lift is conditioned on exposure. Impressions without a click are in its denominator, so the statistic measures the click rate of the user given that the key was shown. It does not measure how often the policy chose the key. Each of the six columns (ℓ^k, f^k) is divided by the standard deviation of its non-zero entries over all user and pool-item cells. The columns are not centred. A pool item with no key value, or a user with no impression on the value, has a zero channel. The channel vector of a pair (u, i) is $z_{ui} \in \mathbb{R}^6$, whose entries are the six standardised statistics of user u at the key values of i: the lift and familiarity for the tag $v_{\text{tag}}(i)$, then for the author $v_{\text{author}}(i)$, then for the music $v_{\text{music}}(i)$. A tilde denotes the standardised statistic. Given the user and the item, the channel is a table lookup.

3.3. LEAF-LGN: channels inside a graph backbone

The backbone is LightGCN (He et al., 2020) on the graph of clicked pre-cut pairs X. The normalisation is symmetric. The model has $d = 64$ dimensions and two propagation layers, and the embeddings used for scoring are the mean of the layer-zero and the propagated embeddings, $\bar{e}_u$ and $\bar{e}_i$. The base score of the implementation is the matrix-factorisation score with biases, so the user and item biases b_u and b_i and a global bias b_0 are learned and are zero-initialised. The biases are not penalised by the weight decay, which applies to the embeddings only. LEAF-LGN scores a pair as

$$s_{ui}^{\mathrm{LEAF}} = \langle \bar{e}_u, \bar{e}_i \rangle + b_u + b_i + b_0 + \beta\, q_i + \sum_{c=1}^{6} g_c\, z_{ui,c}, \tag{4}$$

where q_i is the standardised logarithm $\log(1 + \sum_u X_{ui})$ of the pre-cut click count of i. The scalars β and the six channel gains $g \in \mathbb{R}^6$ are zero-initialised and are learned together with the embeddings and the biases.

The model is trained with the Bayesian personalised ranking (BPR) loss (Rendle et al., 2009). The positives are the clicked pairs X, each with weight 1. The policy channel adds the shown-but-not-clicked pairs N as positives. In every epoch each such pair is kept with probability w, so w is the weight of the policy channel. A negative is an item drawn uniformly from the items outside $X \cup N$ (outside X when $w = 0$), and an item that falls in the excluded set is redrawn once. The loss is the batch mean of $-\log \sigma(s_p - s_n)$ plus a weight-decay term $\mathrm{wd} \sum_{\mathrm{batch}} (\|\bar{e}_u\|^2 + \|\bar{e}_i\|^2 + \|\bar{e}_j\|^2)/|B|$, where j is the negative item. Training uses Adam with batch size 8,192, at most 40 epochs, and early stopping for each split after three epochs without a gain in validation nDCG@10.

In LEAF-LGN the learning rate $\mathrm{lr} \in \{10^{-2}, 3 \cdot 10^{-3}\}$, the weight decay $\mathrm{wd} \in \{10^{-4}, 10^{-5}\}$ and the policy weight $w \in \{0, 0.25, 0.5\}$ define twelve trainings. The grid and the selection rule are given in Section 4. The closed-form variant LEAF-tied, which fits a collective linear autoencoder with the same channels, is reported only in the Stage 1 comparison and in the ablations of the LEAF channels.

3.4. DAX: detection-adjusted attractiveness

DAX models each pre-cut impression of user u on item i in tab t as an examination followed by an attraction:

$$E_r \sim \mathrm{Bernoulli}(\theta_u \kappa_t), \qquad R_r \sim \mathrm{Bernoulli}(\alpha_i), \qquad y_r = E_r R_r. \tag{5}$$

The factor θ_u is the examination probability of the user, κ_t is the examination factor of the scenario, and α_i is the attractiveness of the item. The click probability is therefore $\theta_u \kappa_t \alpha_i$. The scenario factor is normalised by the first tab, $\kappa_1 = 1$. The factors θ and α are probabilities. The factor κ_t is a relative quantity, and its values are not bounded by one (Section 3.6).

Priors.. The attractiveness of an item is shrunk towards the centre c_i of its content group, with prior strength $\nu = 80$ pseudo-impressions. The default centre is the tag centre defined below. The examination of a user is shrunk towards the lifelong prior $\tilde{c}_u$ defined below, with strength $\lambda = 100$ pseudo-impressions. The scenario factor has one pseudo-count in its numerator and one in its denominator, which pulls it towards one for a tab with few impressions.

E-step.. Let $\tau_r = \theta_{u_r} \kappa_{t_r}$. If $y_r = 1$ the event is an examination and an attraction, so $p_{E,r} = p_{R,r} = 1$. If $y_r = 0$, the posterior probabilities are

$$p_{E,r} = \frac{\tau_r (1 - \alpha_{i_r})}{1 - \tau_r \alpha_{i_r}}, \qquad p_{R,r} = \frac{(1 - \tau_r)\, \alpha_{i_r}}{1 - \tau_r \alpha_{i_r}}. \tag{6}$$

M-step.. The three factors are updated in the order θ, κ, α, using the posteriors of the E-step and the most recent value of each factor. Each update is a closed form:

$$\theta_u \leftarrow \frac{\sum_{r:u_r=u} p_{E,r} + \lambda\, \tilde{c}_u}{\sum_{r:u_r=u} \kappa_{t_r} + \lambda}, \tag{7}$$

$$\kappa_t \leftarrow \frac{\sum_{r:t_r=t} p_{E,r} + 1}{\sum_{r:t_r=t} \theta_{u_r} + 1} \quad \text{then} \quad \kappa \leftarrow \kappa / \kappa_1, \tag{8}$$

$$\alpha_i \leftarrow \frac{\sum_{r:i_r=i} p_{R,r} + \nu\, c_i}{n_i + \nu}, \tag{9}$$

where n_i is the number of pre-cut impressions of i. The values are clipped to $[10^{-3}, 1]$ for θ and to $[10^{-4}, 1 - 10^{-4}]$ for α. The factors start at $\theta = 0.7$, $\kappa = 1$ and $\alpha = 0.6$. The EM runs for 40 iterations, with no convergence test. The output is $\operatorname{logit} \alpha$, which is the channel that is combined with LEAF-LGN.

The θ and κ updates are the closed forms that are implemented in `drup/dax.py`. They are not the exact maximisers of the expected complete-data log-likelihood. The exact stationarity condition for θ_u has $(1 - p_{E,r})\kappa_{t_r}/(1 - \tau_r)$ in place of κ_{t_r} in the denominator, and this term is zero for a click. The

implemented update therefore counts every impression in the denominator. The same approximation applies to κ_t. The synthetic check of the observed-data likelihood in `tests/test_dax.py` does not decrease at any iteration when the priors are negligible. We do not claim that the penalised likelihood increases in every iteration of the fits used in the paper.

Tag-mean shrinkage.. The default centre of an item is the pooled attraction of its first tag. For a tag g,

$$c_g = \frac{\sum_{i:\,\mathrm{tag}(i)=g} \sum_{r:\,i_r=i} p_{R,r} + 1}{\sum_{i:\,\mathrm{tag}(i)=g} n_i + 2}, \qquad c_i = c_{\mathrm{tag}(i)}. \tag{10}$$

An item without a tag forms a group of its own. The centre is recomputed at each iteration. The tag centre is the default. The DAX grid of Section 3.5 also contains a Fay-Herriot style alternative (Fay and Herriot, 1979): after the first five iterations, $c_i = \sigma(\mu + x_i^\top \beta)$, where x_i is a set of one-hot content codes (tag, upload type, video type, duration decile, aspect ratio, music type, and three bins of the out-of-pool author history), the ridge penalty is 10, and the weights are $\sqrt{n_i + 1}$. The ablation without a tag prior replaces the centre by one global mean. A user-stratum option follows the MRP idea of (Park et al., 2004): it keeps an attraction $\alpha_{i,s}$ for each stratum s of users (five quantile groups of the lifelong click rate and one group without history), shrunk to α_i with strength 80.

Lifelong user prior.. The centre of the examination prior is the lifelong click rate of the user, rescaled to the examination level of the pre-cut log. Let $\bar{r}_u = r_u / \bar{r}$, where $\bar{r}$ is the mean of r_u over the users of the dataset. The centre is

$$\tilde{c}_u = \mathrm{clip}_{[10^{-3},1]}\Big(\bar{r}_u \, \frac{\sum_r p_{E,r}}{\sum_r \kappa_{t_r}}\Big), \tag{11}$$

where both sums are over the pre-cut impressions at the current iteration. A user without a lifelong history has the mean rate. When $\lambda = 0$ the θ update is $(\sum_{r:u_r=u} p_{E,r} + 1)/(\sum_{r:u_r=u} \kappa_{t_r} + 2)$, so the prior is a weak, uniform one.

3.5. Combination with LEAF-LGN

The DAX channel is standardised over the pool items. Let $\zeta_i = (\mathrm{logit}\,\alpha_i - \mu)/\sigma$, where μ and σ are the mean and standard deviation of $\mathrm{logit}\,\alpha$ over $\mathcal{I}$. For a user u and a candidate list C_u, the combined score is

$$s_{ui} = \mathrm{rstd}_{C_u}\big(s_{ui}^{\mathrm{LEAF}}\big) + \gamma \, \mathrm{rstd}_{C_u}\big(\zeta_i\big), \qquad i \in C_u, \tag{12}$$

where rstd_{C_u} subtracts the mean over the candidate list and divides by its standard deviation. With stratum cells, ζ_i is replaced by the value of the stratum of u. The weight γ takes values in $\{0, 0.05, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5, 0.7, 1.0\}$. At $\gamma = 0$ the score is the backbone score, so the combined model is never worse than the backbone on the selection set.

The combination is post hoc. The backbone is trained alone. At each epoch, every DAX configuration and every γ is scored on validation users. A DAX configuration is one of the 32 settings of the prior strength $\nu \in \{80, 160\}$, the centre (tag or content), the lifelong strength $\lambda \in \{0, 100\}$, the tab factor (off or on), and the stratum cells (off or 80). The selection of each split is the epoch, configuration and γ with the best validation value. The early-stopping rule of each split uses the best validation value over all candidates of the epoch, the backbone and the combined scores alike.

A jointly trained variant is implemented in `drup/dax.py` as `LEAFLGNDAX`. It adds ζ as a seventh channel with a learned weight, and that weight is trained with the BPR objective. No result of this variant is reported in the paper, so the weight γ of the reported combination is a validation-selected hyperparameter and not a learned parameter.

3.6. Identifiability and what the appeal measures

The likelihood of an impression depends on the parameters only through the product $\theta_u \kappa_t \alpha_i$, because $P(y = 1) = \theta_u \kappa_t \alpha_i$ and $P(y = 0) = 1 - \theta_u \kappa_t \alpha_i$. For any constant $c > 0$ the transformation $(\theta, \alpha) \mapsto (c\theta, \alpha/c)$ leaves every term of the likelihood unchanged, as long as the values stay in $[0, 1]$. The normalisation $\kappa_1 = 1$ fixes the scale of the scenario factors. The likelihood therefore identifies the factors up to one global scale that is shared by θ and α. The scale is fixed in practice by the bounds and by the priors, not by the likelihood. We have not shown that no other transformation leaves the likelihood unchanged on the observed design.

The order of α is invariant to this scale, but the logit is not a linear function of the scale. The standardised channel ζ, and therefore the combined score, depends on the prior strengths ν and λ. For this reason these strengths are validation hyperparameters, and the paper does not report the values of α as absolute probabilities.

The factors θ and κ are nuisance parameters. They should not be read as user traits. The scenario factors are relative to tab 1. In the probe fit with prior 80, several tabs have factors above one, which is allowed because the

factors are only identified relative to the reference. Items with few pre-cut impressions are pulled towards their content centre, and users with few clicks are pulled towards their lifelong prior. The median number of pre-cut impressions per pool item is 43 over the 7,583 pool items, with items without a pre-cut impression counted as zero (0.6% of items; `results/v13_data_stats.txt`).

3.7. Cost and measured run times

The LEAF tables are dense user by key-value arrays for the three key families, with one row per user. A pair is a gather, which costs a constant amount per pair. The tables are built in one streaming pass over the lifelong log. The LightGCN propagation costs $O(L|E|d)$ per epoch, where $|E|$ is the number of clicked pairs, and each sampled pair adds $O(6)$ for the channels.

Each DAX iteration is linear in the number of pre-cut impressions plus the numbers of users, items and tabs, because all accumulations are index-add operations. The Fay-Herriot alternative adds one ridge solve per iteration, whose size is the number of covariate levels. The main run measured the following on two threads (`results/logs_v11/dax_lgn_post.log`). It fitted 32 DAX tables in 1.8 to 3.3 s each, which is 77.6 s in total. It trained the twelve LEAF-LGN backbones in 884 to 2,437 s each, which is 19,490 s in total. The run took 19,628 s of wall time with 86,400 validation evaluations and a peak resident memory of 6,032 MB.

3.8. Algorithms

Algorithm 1 gives the DAX fit, and Algorithm 2 gives the training and selection of LEAF-LGN+DAX.

4. Experimental protocol

This section defines the data split in time, the users, the selection rule, the statistical tests, the leakage controls and the baselines. The numbers of the results are in Section 5. Every number in this section is either a count from a data file or a configuration count from a run log, and each is given with its source.

4.1. Temporal protocol and the role of the random-exposure data

The standard log is the policy log of the platform. The random-exposure log is the set of impressions in which a feed slot was replaced by a video drawn uniformly from the pool $\mathcal{I}$. The random-exposure period starts on

Algorithm 1 DAX-EM on the pre-cut log

Require: pre-cut impressions (u_r, i_r, t_r, y_r), tag groups, lifelong rates r_u, prior strengths ν, λ, 40 iterations

$\theta_u \leftarrow 0.7$, $\kappa_t \leftarrow 1$, $\alpha_i \leftarrow 0.6$
for $it = 1, \ldots, 40$ **do**
  for each impression r **do** ▷ E-step
    if $y_r = 1$ **then** $p_{E,r} \leftarrow 1$, $p_{R,r} \leftarrow 1$
    else $\tau_r \leftarrow \theta_{u_r}\kappa_{t_r}$, $p_{E,r} \leftarrow \tau_r(1-\alpha_{i_r})/(1-\tau_r\alpha_{i_r})$, $p_{R,r} \leftarrow (1-\tau_r)\alpha_{i_r}/(1-\tau_r\alpha_{i_r})$
    end if
  end for
  $\tilde{c}_u \leftarrow \mathrm{clip}_{[10^{-3},1]}(\bar{r}_u \sum_r p_{E,r} / \sum_r \kappa_{t_r})$ ▷ lifelong centre
  $\theta_u \leftarrow (\sum_{r:u_r=u} p_{E,r} + \lambda\tilde{c}_u)/(\sum_{r:u_r=u} \kappa_{t_r} + \lambda)$, clipped
  $\kappa_t \leftarrow (\sum_{r:t_r=t} p_{E,r} + 1)/(\sum_{r:t_r=t} \theta_{u_r} + 1)$; $\kappa \leftarrow \kappa/\kappa_1$
  $c_g \leftarrow (\sum_{i\in g} \sum_{r:i_r=i} p_{R,r} + 1)/(\sum_{i\in g} n_i + 2)$ for each tag group g, $c_i \leftarrow c_{\mathrm{tag}(i)}$
  $\alpha_i \leftarrow (\sum_{r:i_r=i} p_{R,r} + \nu c_i)/(n_i + \nu)$, clipped
end for
return $\mathrm{logit}\,\alpha$, θ, κ

2022-04-22 at 00:00 UTC+8, and this instant is the cut-off T_0 of the protocol (Section 3.1).

Every quantity used for training or for selection is computed before T_0. This includes the in-pool click matrix and the standard log used by the backbones, the lifelong tables of Section 3.2, the DAX fit, the popularity offset, and the negative sampler. The training log is the standard log before the cut-off, and the test lists are the random impressions after it. A pair that was shown before T_0 is excluded from its user's candidate list, so a candidate is never a pair that the backbone saw as a positive or as an impression.

The reason for the cut-off is the feedback path. After a random impression the policy keeps logging, so a standard log that extends past T_0 can contain the user's later responses to the randomly inserted video. A protocol that trains on that log can therefore learn from the test outcome. The concurrent protocol, which uses the whole standard log, is reported only as a robustness row in Section 5.

The random-exposure data are used for two purposes only. They provide the candidate lists and labels of the validation users, which select the con-

Algorithm 2 Training and post-hoc selection of LEAF-LGN+DAX

Require: pre-cut log, the twelve backbone settings $(\mathrm{lr}, \mathrm{wd}, w)$, the 32 DAX configurations, the ten values of γ, five splits of the validation and test users

for each backbone setting $(\mathrm{lr}, \mathrm{wd}, w)$ **do**
    train LEAF-LGN (BPR, policy channel w) one epoch at a time, at most 40 epochs
    fit (once per configuration) the DAX table logit α of Algorithm 1
    for each epoch **do**
        compute s_{ui}^{LEAF} on the candidates of every validation and test user
        for each DAX configuration and each γ **do**
            $s_{ui} \leftarrow \mathrm{rstd}(s_{ui}^{\mathrm{LEAF}}) + \gamma\,\mathrm{rstd}(\zeta_i)$ ▷ $\gamma = 0$ is the backbone
            for each split s still active **do**
                if validation nDCG@10 of s_{ui} exceeds the best value of s **then**
                    record the configuration, the epoch, γ, and the test nDCG@10 of s
                end if
            end for
        end for
        each split stops after three epochs without a gain in its best validation value
    end for
end for
return for each split the recorded configuration with the best validation value and its test value

figuration, and of the test users, which measure the score. No model reads the random log for training. This is the setting that matches the uniform-insertion target. A method that trained on uniform labels would be measured on its own training target, and the comparison would no longer test the use of the policy log. Validation labels enter only through the selection rule, and the test labels only through the reported test value.

4.2. Datasets

The primary dataset is KuaiRand-27K (Gao et al., 2022b). Its users are all users that appear in both the standard log and the random log, and its

pool is the set of random-log videos. The secondary dataset is KuaiRand-1K, from the same release with 1,000 users, 993 of whom appear in both logs. Its user identifiers are the same as those of the 27K release, so the 1K results are a replication at a smaller scale and not an independent sample. Table 2 gives the sizes under the temporal protocol.

Coat (Schnabel et al., 2016) and Yahoo! R3 (Marlin and Zemel, 2009) contain ratings that users gave, together with a randomly chosen set of ratings for the test part. They record no impressions that were shown and left unrated, and they have no history on other items. DAX needs impressions with and without a click in each scenario, and LEAF needs a user's history on videos outside the test pool. Neither component can be run on these data, and running the backbones alone would test a different question. KuaiRec (Gao et al., 2022a) has a near-complete small user-video matrix and a larger sparse log of watch ratios. It has no scenario field among its interaction fields, and the user's history is confined to its own matrix, so the history on videos outside the test pool is not available. Its daily item counts are platform aggregates and not user impressions. The Open Bandit Dataset (Saito et al., 2021) has no persistent user identifiers, so no lifelong history can be formed for it.

The motivation for a randomised test is the headroom of the classic benchmarks. A leave-one-user-out item oracle uses test labels, so it is a diagnostic and not a method. On KuaiRec, a training-free log-fitted prior alone reaches nDCG@20 0.6367 on the 1,410 evaluated users, against 0.6378 for DRUP (`v13_saturation_kuairec.txt`, copied from `v8_probes/probe_linear.txt`). On KuaiRand-27K the same kind of diagnostic is in a file, and it is over all users: the item oracle reaches nDCG@10 0.5072, the log prior 0.3884 and the random ranking 0.3329. A rank-64 model on the residuals of the uniform item means moves the item-mean value from 0.5850 to 0.5851 (`v9_probes/headroom_kuairand_pure.txt`). The residual headroom therefore lies at the item level, and that is the level at which DAX operates.

4.3. Splits, users and per-split selection

Each split partitions the users of the test list into validation and test users, by user and not by impression. The split with index $s \in \{0, \ldots, 4\}$ takes a random permutation of the users with seed s, and puts the first $\mathrm{round}(0.3\,m)$ users into validation. For KuaiRand-27K this gives $\mathrm{round}(0.3 \times 27{,}077) = 8{,}123$ validation users and 18,954 test users in each split. For KuaiRand-1K the first split has 298 validation and 695 test users (`v9_probes/stats_`

`1k.txt`). The two parts share no user, and the code asserts this for every split (`drup/leaf.py protocol_split`). Users with no click among their random impressions have no defined nDCG and are not evaluated, so the evaluated test users per split are the numbers in Table 2.

The training runs do not depend on the split. Each backbone is trained once for each hyperparameter setting on the whole pre-cut log, and every split reads the same training trajectory. The splits change only which users select the epoch, the configuration and the weight γ, and which users are scored. This is why the split-level test of Section 4.4 is needed.

For each split, the selected configuration is the one with the largest validation nDCG@10. The configuration contains the backbone setting, the epoch, the DAX configuration and γ where they apply. The test labels are read only for the configuration that is the best on validation so far, and the value stored for the split is the value of the final best configuration. No test label influences the choice of any configuration.

The configuration count of a method, reported as n_{configs} in the run logs, is the number of backbone trainings times the number of DAX configurations times the number of γ values. For a backbone without DAX it is the number of trainings or hyperparameter settings. Epochs are selected inside each training but are not counted as configurations. They are counted in the number of evaluations n_{evals} instead. For example, LEAF-LGN+DAX has $12 \times 32 \times 10 = 3{,}840$ configurations and 86,400 evaluations, and QP family+DAX has $10 \times 32 \times 10 = 3{,}200$ configurations. Table 4 gives these counts for every method, and the search budgets differ. The fair baseline is the one with the same DAX module.

4.4. Statistical tests

Paired per-user differences.. For two methods A and B, a user's difference is the mean, over the splits in which the user is evaluated, of the difference of the two nDCG@10 values. The reported quantity is the mean of these user differences, multiplied by 10^3, with a 95% half-width from the two-sided t interval over users. The p-value is that of a two-sided one-sample t test on the user differences. Users appear in several test parts and share one training run, so the user-level t test is approximate, and it is reported together with the split-level test.

Split-level test.. The Nadeau-Bengio corrected resampled t test (Nadeau and Bengio, 2003) is applied to the five split means $\bar{d}_s$ of the difference $d_s =$

$\mathrm{nDCG}_A(s) - \mathrm{nDCG}_B(s)$. With $J = 5$ splits, the variance of the mean is estimated as

$$\widehat{\mathrm{Var}} = s_d^2\Big(\frac{1}{J} + \frac{n_{\mathrm{test}}}{n_{\mathrm{val}}}\Big) = s_d^2\Big(\frac{1}{5} + \frac{7}{3}\Big), \tag{13}$$

where s_d^2 is the sample variance of the J split differences and the ratio is 0.7/0.3. The statistic is $t = \bar{d}/\sqrt{\widehat{\mathrm{Var}}}$, compared with a t distribution with $J-1 = 4$ degrees of freedom. We take the validation part as the training part of the original correction and the test part as its test part. The ratio 7/3 is larger than the alternative 3/7, so this choice inflates the variance more and is the more conservative of the two. With these settings, the two-sided test at level 0.05 needs a mean-to-standard-deviation ratio of the split differences of at least $t_{0.975,4}\sqrt{1/5 + 7/3} \approx 4.42$.

Holm adjustment.. The p-values of a family are adjusted by the step-down procedure of Holm (1979). Each table names its family. The primary family for gate B1 has size five and was fixed before the Stage B runs: LEAF-LGN+DAX against the QP family, the QP family with DAX, the post-hoc LEAF-LGN backbone, r-AdjNorm and LightGCN. The table family of the deep-baseline extension contains every comparator that is present in the table, and it was defined after the Stage B results. It has seven members with the deep baselines. The Stage 1 family for the closed-form comparison contains the two LEAF variants against four baselines, so it has eight members. The same adjustment is applied to the t p-values and to the Nadeau-Bengio p-values, separately.

Gate and reading.. The gate B1 rule was written in `paper/v7/DESIGN.md` (Section 7, dated 2026-11-08) before the Stage B runs. It requires that LEAF-LGN+DAX exceed the QP family by at least 15×10^{-3} per user, and that both Holm-adjusted p-values be below 0.05. The comparison with the fair baseline, QP family with DAX, has no pre-set margin. It is read with the same significance level. We report the t and Nadeau-Bengio p-values only. No Wilcoxon test is used.

4.5. Leakage checks and controls

Table 3 lists every test in `tests/` and the checks printed by the runs and the controls of the analysis. The unit tests were run on 2026-11-13 and all 15 of them passed (6 in `tests/test_lifelong.py` and 9 in `tests/test_dax.py`). The kind of each number is given in the table. A "main" number is

a validation-selected result of the main setting, a "probe" number is a post-hoc value on stored scores, a "diagnostic" number uses random-exposure labels of the validation users and is never a method or a selection input, and a "code" or "unit" row states what a test or a code path asserts.

4.6. Baselines

Table 4 lists the baselines, their grids and their configuration counts. All baselines are trained on the same pre-cut log with the same splits and the same selection rule. Every backbone with DAX gets the same post-hoc DAX module and the same γ grid as LEAF-LGN.

The QP family is our own earlier method. It is not cited as a publication, and its full description is in Appendix A. The QP family searches 20 configurations in total (4, 8 and 8 for its three members), and each split selects one of them on validation, as described above.

The DIN and SIM baselines are lifelong-aware. Their sequence is the last $L = 200$ pre-cut events of the user on videos outside the pool, with the tag, author and music codes of each video and an engagement sign. DIN attends over the whole sequence. SIM-hard first keeps the $K = 50$ most recent events that share the tag or author of the candidate, and then attends over them. Both models also use the identifier embeddings of users and items and are trained with the BPR loss on the pre-cut clicks, as LightGCN is. Their grids are smaller than those of the graph backbones, with dimension 16 instead of 64, at most 12 epochs instead of 40, and patience 2 instead of 3. The disclosure matters for the comparison: the DIN and SIM budgets are four trainings, and the results on them are conditional on that budget. The SIM run on KuaiRand-27K did not finish. Its partial file holds three of the four trainings and no SIM result for that setting is reported. SIM is therefore reported on KuaiRand-1K only.

The NAVIP method (Kim et al., 2022) is not re-run. It changes the neighbour aggregation of a graph network and does not use lifelong histories, and no run of it is reported. TWIN (Chang et al., 2023) and the soft-search variant of SIM are not run, and the comparisons in the paper do not include them.

4.7. Ablation design

The ablations of DAX are a cut-down version of the main search. Each ablation arm is a separate method and is selected per split in the same way as the full arm. The arms are run on the backbone settings that the main run

chose: each split reads the epochs within three epochs of its chosen epoch, and the trainings are the two settings that the splits chose. The arms use five values of γ, $\{0.1, 0.25, 0.5, 0.7, 1.0\}$, instead of the ten of the main run. The arms are the three DAX switches (no tab factor, no lifelong prior, no tag prior; each with 16 configurations), the additive MRP-style logistic with tab effects (32 configurations, matched to the 32 of the full grid), and the two permutations of the attractiveness: within tag (the tag mean is kept) and across all items. Each permutation arm is run with three permutation seeds.

Because the trainings and epoch windows are restricted, the base of the ablation run is its own LEAF-LGN, which is selected inside the window and has a lower value than the main-run LEAF-LGN. The ablation gains are paired differences against that base, so they are not the same quantity as the main-table DAX gain against the main-run backbone. The full arm of the ablation run selects the same configurations on all five splits as the main run and reproduces its test value exactly (Table 3).

The gate B2 conditions are fixed in `paper/v7/DESIGN.md` (Section 7): the within-tag permutation must lower the gain by at least 30%, the across-item permutation must remove at least 90% of it, and the additive arm must stay at least 4×10^{-3} below the full DAX arm. The stored analysis reports the raw paired p-values of the ablation arms and does not adjust them. The Holm adjustment within the ablation family is therefore not part of the reported results, and the gate decisions do not use p-values.

4.8. Kinds of number

Three kinds of number are used, and they are not mixed within one table without a label. A main number is a validation-selected result of the main setting, from the files in `results/v10/` and `results/v11/`, and of the v12 deep baselines. A probe number is a post-hoc computation on stored scores, from `results/v11_probes/` and `results/v9_probes/`. A diagnostic number uses random-exposure labels of the validation users and is never a method or a selection input. A number that is not in a result file is marked as TBD in the text, and the table entry names the file that would supply it.

5. Results

All numbers in this section are produced by `experiments/v13_tables.py` and `experiments/v13_figs.py` from the result files named in each table

note. Differences are per-user paired differences of nDCG@10 multiplied by 10^3, with the 95% half-width in brackets. A comparison is called significant only when the Holm-adjusted p value is below 0.05 and the test is named. Tests are the per-user paired t test over the pooled test users (n = 22,612 on KuaiRand-27K) and the Nadeau-Bengio corrected resampled t test over the five split means (Nadeau and Bengio, 2003); the Holm family is stated in each table.

5.1. Methods and search budgets

Table 5 lists the mean and standard deviation over the five splits. The search budgets differ by two orders of magnitude, so they are reported with every number. LEAF-LGN+DAX searches 3,840 configurations per split, the QP family with DAX 3,200, and the QP family itself 20. The budget of the fair baseline is therefore not matched in the strict sense; Section 5.3 discusses what this does to the comparison.

5.2. Main comparison

Table 6 gives the paired comparisons of LEAF-LGN+DAX with each comparator of the main family. The family has seven members: the QP family, the QP family with DAX, the LEAF-LGN backbone without DAX, r-AdjNorm, LightGCN, DIN and DIN with DAX. SIM and SIM with DAX are part of the family but have no result file for KuaiRand-27K yet, so their rows are marked as pending and the Holm correction runs over the seven comparators present. EASE and EASE with DAX are reported outside the family. Table 7 shows that adding them to the family does not change any conclusion at the 0.05 level.

The per-user paired test gives an adjusted p value below 10^{-11} for each of the seven comparators, and the sign is the same in every split for each of them (Figure 1). The split-level test is also below 0.05 after Holm adjustment for all seven comparators, but the margins are small for the two QP comparators and for DIN with DAX, whose adjusted Nadeau-Bengio p values are 4.56×10^{-2}, 2.23×10^{-2} and 4.56×10^{-2}. With five splits the split-level test has low power, and these values are the closest to the threshold in the paper.

The pre-declared gate for the primary comparison requires a paired difference of at least 15×10^{-3} against the QP family, with both adjusted p values below 0.05 (`results/v11/stageB.txt`, Section 3). The observed difference is +22.32 [1.83], and the gate is passed. The gate was declared before

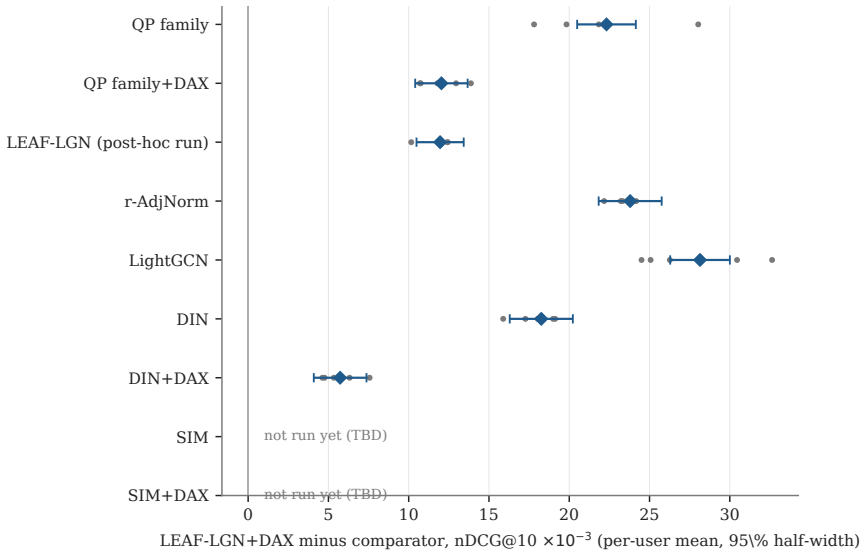


Figure 1: Paired differences of LEAF-LGN+DAX against each comparator of the main family. Grey dots: the five split differences. Diamond: pooled per-user mean with its 95% half-width. Source: `results/v11/dax_lgn_post.json` and the comparator files named in Table 5.

Stage B, and the combination it evaluates is the post-hoc one described in the method section.

5.3. Fair-baseline reading

The QP family has the smaller search budget, so the fairer reference for the DAX module is the QP family with DAX, which searches the QP backbones selected without DAX. Against that baseline LEAF-LGN+DAX is higher by +12.04 [1.63] (per-user t test, Holm-adjusted $p = 7.92 \times 10^{-47}$; Nadeau-Bengio Holm-adjusted $p = 2.23 \times 10^{-2}$). The DAX increment on the QP backbone is +10.28 [1.37] with an unadjusted Nadeau-Bengio p of 0.105, so on the split-level test the increment on the QP backbone alone is not significant. The LEAF backbone without DAX, against the stored QP family, is +9.63 [1.60] with an unadjusted Nadeau-Bengio p of 0.063; in the stage 1 run this comparison is not significant after Holm adjustment (Table 8). The LEAF backbone is therefore not, by itself, a clear improvement over the QP family on the split-level test; the combination is.

5.4. DAX on other backbones

Table 9 applies the same post-hoc DAX module to four backbones with a result file. The increments are +11.98 [1.64] for EASE (unadjusted Nadeau-Bengio p 0.000), +10.28 [1.37] for the QP family, +11.96 [1.47] for LEAF-LGN and +12.53 [1.41] for DIN. The LightGCN increment is pending (`results/v11/dax_lightgcn.json` is missing), and so is the SIM increment. The increment is positive on every backbone with a result, which is the basis for describing the module as backbone-agnostic. The claim is about the module's increment under the same post-hoc rule, not about the backbones' absolute accuracy.

5.5. Ablations of DAX

Table 10 decomposes the DAX module. The arms are run in a reduced setting: two backbone trainings, five gammas and an epoch window of three around the main selection. The full arm reproduces the per-split values of the main run, so the arms can be compared with each other. The gain of each arm is measured against the LEAF-LGN base of that run, which has a mean of 0.4706; this base is lower than the 0.4736 of the main run, so the gains in Table 10 are larger than the main increment of +11.96 [1.47]. The comparisons that do not depend on the base are the differences from the full arm, in the rightmost columns.

Each removal lowers the score. Removing the examination factor κ_t costs +1.51 [0.79] (Holm-adjusted $p = 2.02 \times 10^{-4}$ by the per-user t test), removing the lifelong prior on θ_u costs +1.78 [0.61] ($p_{\text{Holm}} = 2.16 \times 10^{-8}$), and removing the tag prior on α_i costs +3.41 [0.99] ($p_{\text{Holm}} = 4.79 \times 10^{-11}$). Replacing the multiplicative form by the additive logit with tab effects costs +6.09 [0.96], which is the largest of the structural comparisons ($p_{\text{Holm}} = 5.43 \times 10^{-35}$); the additive arm keeps 59.7% of the full gain.

The shuffle controls test the item-level part of α_i. Permuting α across all items removes the whole gain: the pooled shuffled arm has a gain of −0.27 [0.36], that is −1.8% of the full gain, and the full arm is higher by +15.36 [1.40] ($p_{\text{Holm}} = 3.53 \times 10^{-101}$). Permuting α within each tag keeps 41.6% of the full gain (+6.28 [1.10]), and the full arm is higher by +8.81 [1.00] ($p_{\text{Holm}} = 2.47 \times 10^{-65}$). The within-tag arm therefore keeps a substantial part of the gain, which is the tag mean, and the remaining part lies in the item-level deviation from it. The three permutation seeds give the same ordering (Table 10, lower block): the across-item shuffles have gains between −0.44 and −0.16 and the within-tag shuffles between +5.66 and +6.84.

5.6. KuaiRand-1K replication

Table 11 repeats the primary comparisons on KuaiRand-1K, which contains 842 test users pooled over the five splits. LEAF-LGN+DAX reaches 0.4590 against 0.4246 for the QP family, a difference of +37.58 [11.10] (per-user t test, Holm-adjusted $p = 1.61 \times 10^{-10}$; Nadeau-Bengio Holm-adjusted $p = 2.28 \times 10^{-2}$), and against 0.4358 for the QP family with DAX, a difference of +25.82 [9.66] (Holm-adjusted $p = 3.91 \times 10^{-7}$; Nadeau-Bengio Holm-adjusted $p = 2.18 \times 10^{-2}$). The DAX increment on the QP backbone, +11.76 [8.54], is significant under the per-user t test (Holm-adjusted $p = 7.04 \times 10^{-3}$) but not under the split-level test (Nadeau-Bengio Holm-adjusted $p = 0.160$). The 1K replication uses a wider QP search for the DAX arm (6,400 configurations against 3,200 at 27K), which favours the baseline. The 1K users are a subset of the 27K users, so the replication has lower power and is not an independent sample.

5.7. Training seeds

Table 12 repeats the LEAF-LGN+DAX run with training seeds 1 to 3. The QP family was not re-trained with these seeds, so no seed comparison with it is reported. The LEAF-LGN+DAX mean is 0.4854 for seed 0, 0.4858, 0.4856 and 0.4844, with a spread over seeds of 0.4853 ± 0.0006. The within-seed gain over LEAF-LGN is positive in all four seeds, between +11.96 and +15.10, with per-user t p values below 10^{-56} and Nadeau-Bengio p values of at most 0.002. These seed tests are not in a Holm family and are not called significant; they show that the gain is not a feature of one training run. Seeds 1 to 3 use a reduced backbone grid, as the table note states, which limits what the seed spread can show about the full search.

5.8. Lifelong deep baselines

DIN attends over the last 200 lifelong events of each user. On KuaiRand-27K it reaches 0.4675 and is below LEAF-LGN+DAX by +18.26 [1.97] (per-user t test, Holm-adjusted $p = 5.62 \times 10^{-73}$). DIN with DAX reaches 0.4796 and remains below LEAF-LGN+DAX by +5.73 [1.64] (Holm-adjusted $p = 7.95 \times 10^{-12}$; Nadeau-Bengio Holm-adjusted $p = 4.56 \times 10^{-2}$). SIM-hard, which keeps the 50 most recent events that share the candidate's tag or author, has no KuaiRand-27K result file yet, so its rows are pending. On KuaiRand-1K, LEAF-LGN+DAX (0.4590) is above DIN with DAX (0.4228) and above SIM-hard (0.4246) and SIM with DAX (0.4357); the unadjusted differences are reported in Table 13. The deep models are tuned over four

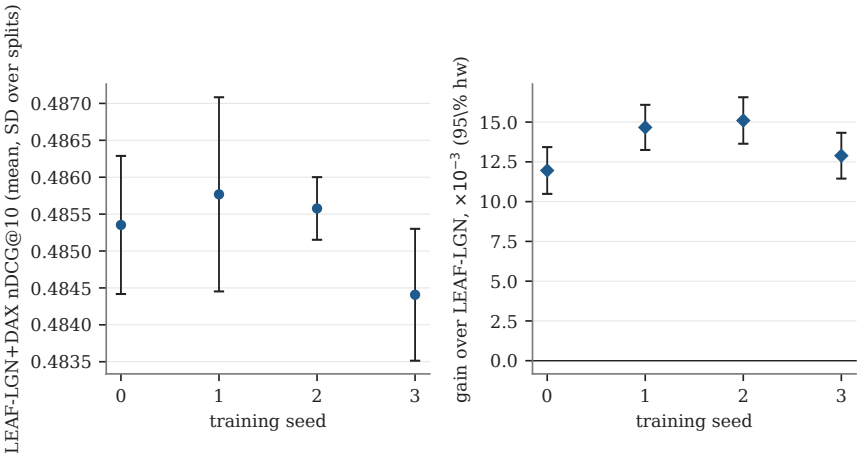


Figure 2: Training seeds 0 to 3 on KuaiRand-27K. Left: LEAF-LGN+DAX mean nDCG@10 with the standard deviation over splits. Right: within-seed gain over LEAF-LGN with the 95% half-width. Source: `results/v11/dax_lgn_post.json` (seed 0) and `results/v11/c_seeds/lgn_s1.json` to `lgn_s3.json`.

configurations each (learning rate and weight decay, with the DAX arms on the post-hoc grid), so they are not tuned as widely as LEAF-LGN. The paper does not claim that LEAF-LGN+DAX is better than a lifelong deep model that has been tuned over a wider grid.

5.9. What the controls show

The LEAF controls come from the stage 1 run on the LEAF-tied backbone, the closed-form variant on which those ablations were run (Table 14). Shuffling the users' lifelong tables across users gives the largest loss, +18.90 [1.99] (Holm-adjusted $p = 4.90 \times 10^{-76}$; Nadeau-Bengio Holm-adjusted $p = 4.60 \times 10^{-3}$), and removing all lifelong channels gives +17.88 [2.04]. The design check of that run, the difference between the shuffled-user arm and the no-lifelong arm, is -1.02×10^{-3}, within its $\pm 2 \times 10^{-3}$ tolerance. The policy channel contributes +5.01 [1.20] (per-user t test Holm-adjusted $p = 6.91 \times 10^{-16}$), but its split-level test is not significant after Holm adjustment (Nadeau-Bengio Holm-adjusted $p = 0.132$). The LEAF-LGN ablations have no result file (`results/v10/leaf_lgn_ablate.json` is missing), so the controls are reported on the LEAF-tied backbone only.

5.10. Summary of the evidence

Across the controls the picture is consistent. The LEAF signal depends on the users' lifelong histories and not only on the item channels: shuffling the users removes the gain. The DAX module depends on the item-level appeal beyond the tag mean and on its user and scenario terms, and a shuffle across items removes its gain. The combination is above every comparator with a result, including the lifelong deep models with DAX on KuaiRand-27K and KuaiRand-1K, and its gain is not specific to one training seed or to the 1K subset. The controls do not establish that the item-level estimate is an unbiased measure of item quality; Section 7 lists what remains untested.

6. Discussion

Why the multiplicative form.. The additive alternative keeps the same covariates and tab effects, but replaces the product of user examination and item attractiveness by an additive logit. Against the full DAX arm it is lower by +6.09 [0.96] (per-user t test, Holm-adjusted $p = 5.43 \times 10^{-35}$), and it keeps 59.7% of the full gain (Table 10). Under the generative model of the method section, a non-click is weak evidence against an item for a user who rarely clicks anything, and weak evidence in a scenario that rarely yields clicks. The multiplicative form encodes that dependence; the additive form, once the user effect is absorbed, treats the two cases alike. The comparison is a fit comparison under one protocol, and it does not show that the multiplicative form is correct in an absolute sense.

What the item-level part contributes.. Permuting α within tags keeps 41.6% of the gain, while permuting it across items removes all of it (Table 10). A model that used only a tag mean would therefore capture part of the signal, and the remainder is carried by item deviations from that mean. This is the reason the item-level estimate is retained, but the decomposition is only as strong as the permutation controls, which keep the marginal distribution of α fixed.

Why the lifelong channels are not interchangeable with the item channels.. Shuffling the lifelong tables across users removes the LEAF gain, and removing all lifelong channels removes +17.88 [2.04] (Table 14). The policy channel contributes less (+5.01 [1.20]) and its split-level test is not significant after Holm adjustment. The signal that transfers to randomly exposed items is

therefore mainly the users' own lifelong affinities for content keys, and the in-pool policy channel adds less on top of it.

Complementarity rather than stand-alone accuracy.. DAX is used as an increment on top of a backbone, and the tables report only that increment. The increment is positive on every backbone with a result (Table 9), and the combination is above the backbones on which it is added. The paper does not report DAX as a stand-alone ranker: the claim is that its score carries information that the backbone scores do not, under the post-hoc rule.

Lifelong deep models.. DIN with DAX is the closest comparator in the main family. LEAF-LGN with DAX is above it by +5.73 [1.64] with a Holm-adjusted Nadeau-Bengio p of 4.56×10^{-2}, which ties with the QP family as the weakest split-level margin of the seven comparators. The lifelong deep baselines are tuned over four configurations each. Whether SIM with DAX, once run, changes the ordering is open, and the paper makes no claim about it.

What the remaining headroom would require.. The gains are concentrated in the item-level appeal estimate and in the lifelong user channels. Further gains would need a different source of information about item quality for random viewers, such as richer content for items or a direct measurement of uniform-exposure responses, and not a larger search over the present hyperparameters. This is an interpretation of the controls and not a tested claim.

7. Limitations

The limitations below are stated in the order in which they bear on the claims of the paper.

One platform family.. KuaiRand-27K and KuaiRand-1K come from one platform, one period and overlapping users. We found no other public dataset that combines randomised insertion, impression logs and lifelong history of the same users, so the evidence describes one setting. The 1K users are a subset of the 27K users and are not an independent replication.

Few splits and a low-power split-level test.. The main comparisons use five user-level splits. The Nadeau-Bengio test over five split means has low power, and the adjusted split-level p values against the two QP comparators and against DIN with DAX are 4.56×10^{-2}, 2.23×10^{-2} and 4.56×10^{-2} in

Table 6. The per-user t test is far below these values, but it treats the users as independent units across splits, which the Nadeau-Bengio test does not assume.

The novelty is a combination.. Each component has a precedent: lifelong behaviour models, examination models in click modelling, and shrinkage towards a group mean. The contribution is the identification of which log-only signals transfer to a uniform-insertion target, the factorisation of examination into a user and a scenario term, and the controlled evidence in Section 5. A reviewer may describe DAX as a click model applied to items; the paper does not dispute that description.

The combination is post hoc.. The mixing weight γ is a validation-selected hyperparameter, and the DAX module is not trained jointly with the backbone in the reported results. The jointly trained variant has no result file (`results/v11/dax_lgn_joint_small.json` and `dax_lgn_joint.json` are missing), so no number for it is reported, and the DAX weight is not described as learned.

Unequal search budgets and grid edges.. LEAF-LGN+DAX searches 3,840 configurations per split and the QP family searches 20, a difference of two orders of magnitude. The QP family with DAX (3,200) reduces the imbalance but searches only the QP backbones selected without DAX, which is conservative for the method and not neutral for the baseline. On KuaiRand-1K the QP family with DAX searches 6,400 configurations, which favours the baseline there. Two DAX choices lie at the edges of their grids: the prior strength is selected at its lower edge 80 (of $\{80, 160\}$) and the lifelong strength k_{life} at its upper edge 100 (of $\{0, 100\}$) in the main run (Appendix Appendix C). The extended grid run that would show whether the optimum lies outside the default grid (`results/v12/daxwide_lgn.json`) has no result file, so this is an open question.

Identifiability up to scale.. The examination and attractiveness factors are identified only up to a scale shared by θ_u and α_i. The order of α_i is unaffected, but the logit transform used in the combination is not invariant to the scale, so the prior strengths are hyperparameters selected on validation. The examination factors θ_u and κ_t are nuisance parameters and are not interpretable as absolute user or scenario traits.

Reduced grids in the ablations and the seeds.. The DAX ablations use two backbone trainings, five gammas and an epoch window of three around the main selection. Their base is the LEAF-LGN of that reduced run (mean 0.4706), not the base of the main run (0.4736), so the gains against the base are not comparable with the main increment. The comparisons against the full arm do not depend on the base, and the full arm reproduces the main per-split values. Training seeds 1 to 3 use a fixed backbone (learning rate 3e-3, weight decay 1e-4) with two trainings in the DAX arm.

Seeds for the QP family are not run.. The seed replicates cover LEAF-LGN+DAX only. The QP family has a partial, incomplete seed run (`results/v11/c_seeds/qp_s1.json.partial.json`) that is not used. The within-seed gains are therefore not compared with the QP family under varying seeds, and the seed table reports the LEAF-LGN gains only.

Deep lifelong baselines are tuned narrowly.. DIN and SIM-hard are each tuned over four configurations (learning rate and weight decay), with the DAX arms on the post-hoc grid. SIM-hard with DAX has no KuaiRand-27K result file, so the comparison with SIM at the full scale is open. The claim that LEAF-LGN+DAX is better than these lifelong deep models is therefore limited to the tuning reported here. A deep model tuned over a wider grid, or with a sequence length other than 200 for DIN, may close the gap; we did not test this.

Single training seed per split in the main table.. The main table uses one training seed per split. Seeds 1 to 3 for LEAF-LGN+DAX support the direction of the gain but are not a full variance estimate.

Click label only.. The main table uses the click label. A long-view label is not reported, because its result file (`results/v11/c_longview*.json`) does not exist. Concurrent-protocol results for the full model are also missing (`results/v11/c_concurrent.json`); the concurrent protocol is a separate robustness setting and is not claimed here.

Remaining headroom.. The combination does not close the item-level gap of the uniform-rate oracle, and the diagnostic in Section 2 reports the remaining headroom. The paper does not claim that DAX is the last step towards that bound.

Coarse content.. KuaiRand ships no titles, covers or frames. The tag is the main content key for LEAF and DAX, and the content centre of DAX is a ridge regression on coarse codes. Results with richer content would be needed before the content channels are judged more widely.

Test scenario and tab factors.. The test scenario is tab 1, and its examination factor is anchored at $\kappa_1 = 1$. The transfer of the tab factors to other scenarios is not tested.

Item-level versus tag-level.. The within-tag shuffle keeps 41.6% of the gain and the across-item shuffle removes it (Table 10). These are permutation controls on a fitted model, not a fitted item-level model with a separate tag-level counterpart, so they show that the item deviations from the tag mean carry signal under this protocol, and nothing more.

Scope of the claims.. The paper does not claim unbiasedness, causal identification of item quality, generality beyond one platform family, or any comparison with published numbers from other protocols, which use different binarisation, candidate sets and selection rules.

8. Conclusion

When a graph recommender is evaluated on items inserted uniformly at random into feeds, two log-only signals help it rank those items. Lifelong, exposure-conditional affinities of users for content keys (LEAF) transfer from items outside the test pool through content keys only, and the gain disappears when the users' lifelong tables are shuffled. A detection-adjusted item attractiveness (DAX), fitted by EM to a click model that discounts non-clicks by user and scenario examination, adds a further increment on top of EASE, the QP family, DIN and LEAF-LGN. The increment vanishes when attractiveness is shuffled across items, while within-tag shuffles keep part of it.

On KuaiRand-27K, under a temporal protocol with five user-level splits, LEAF-LGN with DAX is higher than each of the seven comparators of the main family that have results, with Holm-adjusted per-user t tests below 10^{-11} and split-level Nadeau-Bengio Holm-adjusted p values of at most 4.56×10^{-2}. The comparison with the QP family with DAX, which is the fair baseline in this study, gives +12.04 [1.63]. The result holds on KuaiRand-1K and across three training seeds for LEAF-LGN with DAX.

The evidence is limited to one platform family, five splits, and search budgets that differ by two orders of magnitude. The claims are the controlled ranking gain under the stated protocol and the module's increment on several backbones. The paper does not claim unbiasedness, causal identification of item quality, or superiority over lifelong deep models tuned more widely, and the SIM comparison remains to be run.

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Appendix A. The QP family

The QP family is an earlier method of the authors. It has not been published, and it is described here in full so that the baseline can be reproduced from this appendix. It is a weighted Bayesian personalised ranking (BPR) objective (Rendle et al., 2009) on a LightGCN backbone, with three variants, DR-BPR-LGN-1, DR-BPR-LGN-p2 and DR-BPR-LGN-p. The variants differ only in an offset term, and each is trained on the same pre-cut in-pool log as LEAF-LGN.

Exposure model.. The probability P_{ui} that item i was exposed to user u comes from a logistic exposure model on item popularity (the `pop` option of `drup/estimation.py`, `get_propensity`). The propensities are clipped below at 0.05.

Imputation.. The outcome imputation $\hat{Y}_{ui}$ is the additive model of `baseline_imputation` with inverse-propensity weights, clipped to $[0, 1]$ for the weights and to $[0.01, 0.99]$ for the offset. Its standardised logit z_{ui} is the offset feature.

Unlogged-pair weight.. For a pair that is not a logged positive, $Q_{ui} = \hat{Y}_{ui}(1 - P_{ui})/(1 - P_{ui}\hat{Y}_{ui})$, clipped to $[0, 1]$. It is the probability that such a pair is a positive after all, under Y independent of exposure given the imputation.

Loss.. For a logged positive pair (u, i) and a sampled negative j (resampled when (u, j) is a logged positive), the weight of the negative is $w_{uj} = \max(1 - Q_{uj}, 10^{-3})$ and the loss is

$$\mathcal{L} = -\frac{\sum_{(u,i,j)} w_{uj} \log \sigma(s_{ui} - s_{uj})}{\sum_{(u,i,j)} w_{uj}},$$

with the score $s_{ui} = \langle e_u, e_i \rangle + \beta\, z_{ui}$. The DR-BPR-LGN-1 variant has $\beta = 0$. The DR-BPR-LGN-p2 and DR-BPR-LGN-p variants use $\beta \in \{1, 2\}$ and $\beta \in \{0.5, 1\}$ respectively.

Grids and selection.. The backbone is LightGCN with two layers and dimension 64. The grid is learning rate in $\{10^{-2}, 3 \times 10^{-3}\}$ and weight decay in $\{10^{-5}, 10^{-4}\}$, with the offset grid above. That gives 4 configurations for DR-BPR-LGN-1, 8 for DR-BPR-LGN-p2 and 8 for DR-BPR-LGN-p, 20 in total (`results/v10/learned_kuairand_t_qp.json`). Training runs for at most 40 epochs with patience 3, and batch size 8,192. Validation nDCG@10 selects the configuration in each split. The QP family in the main comparison is the variant with the largest validation value in each split, so its search is 20 configurations per split. The QP family with DAX (the method section) reuses the QP backbones selected without DAX.

Controls on the same family.. The variants with the weight replaced by a popularity weight, by the imputation alone, or by a shuffled weight are defined in `experiments/run_learned.py` and are not part of the reported results.

Appendix B. EM for detection-adjusted attractiveness

This appendix gives the model and the updates implemented in `drup/dax.py` (`fit_dax`). Only rows with time before the cut-off T_0 are read.

Model.. A logged impression of user u, item i and tab t has an examination indicator E and an attractiveness indicator R,

$$E \sim \text{Bernoulli}(\theta_u \kappa_t), \qquad R \sim \text{Bernoulli}(\alpha_i), \qquad y = ER,$$

where y is the click. The factors are in $[0, 1]$, $\kappa_1 = 1$ for the tab with code 1, which is the test scenario.

Priors.. The priors are $\alpha_i \sim \text{Beta}(a\, c_i,\ a(1-c_i))$ with a the prior strength (80 in the main run), $\theta_u \sim \text{Beta}(k\, \tilde{r}_u,\ k(1-\tilde{r}_u))$ with k the lifelong strength (100 in the main run), and κ_t free. The lifelong rate $\tilde{r}_u$ is the user's lifelong click rate rescaled to have mean one and multiplied by the pooled examination rate of the pre-cut log, and is clipped to $[10^{-3}, 1]$. With $k = 0$ the code uses $\text{Beta}(2, 2)$ on θ_u. The attractiveness centre c_i is the pooled attraction of the tag of i with a $\text{Beta}(1, 1)$ smoothing, or, in the content variant, a weighted ridge regression (penalty 10) of logit α on one-hot content codes with weights $\sqrt{n_i + 1}$; the content variant replaces the tag centre after five iterations.

Initialisation and iterations.. $\theta_u = 0.7$, $\kappa_t = 1$, $\alpha_i = 0.6$. There are 40 EM iterations, and no convergence test is applied.

E-step.. Let $\tau = \theta_u \kappa_t$. A click gives $P(E = 1) = P(R = 1) = 1$. A non-click gives

$$P(E = 1 \mid y = 0) = \frac{\tau(1 - \alpha_i)}{1 - \tau\alpha_i}, \qquad P(R = 1 \mid y = 0) = \frac{(1 - \tau)\alpha_i}{1 - \tau\alpha_i}.$$

M-step.. Let $S_u^E = \sum P(E = 1)$ over the rows of u, $K_u = \sum \kappa_t$ over the same rows and $c_u^0 = \tilde{r}_u \sum_u S_u^E / \sum_u K_u$, clipped to $[10^{-3}, 1]$. Then

$$\theta_u = \frac{S_u^E + k\, c_u^0}{K_u + k} \quad (k > 0), \qquad \theta_u = \frac{S_u^E + 1}{K_u + 2} \quad (k = 0),$$

clipped to $[10^{-3}, 1]$. For the tab factors, with κ of the tab coded t,

$$\kappa_t = \frac{\sum_{\text{rows of } t} P(E = 1) + 1}{\sum_{\text{rows of } t} \theta_u + 1},$$

and all κ_t are divided by the value of the reference tab (code 1) after each update, so that it equals one. Note that the code adds a pseudo-count of one to the numerator and denominator; this smoothing is part of the reported model. The attractiveness update is

$$\alpha_i = \frac{\sum_{\text{rows of } i} P(R = 1) + a\, c_i}{n_i + a},$$

clipped to $[10^{-4}, 1 - 10^{-4}]$, where n_i is the number of pre-cut rows of item i.

Strata.. The optional user-stratum cells $\alpha_{i,s}$ of the main run use the same update with α_i as the centre and the cell prior strength `prior_s` $= 80$ (`results/v11/dax_lgn_post.json`, chosen configurations).

Output.. The fit returns $\operatorname{logit} \alpha_i$, θ_u and κ_t. The combined score of the method section uses $\operatorname{logit} \alpha_i$ standardised over each user's candidate list.

Appendix C. Grids and chosen configurations

The search spaces are as follows. LEAF-LGN and the other post-hoc backbones use learning rate $\{10^{-2}, 3 \times 10^{-3}\}$ and weight decay $\{10^{-4}, 10^{-5}\}$,

with the backbone's own grid. LEAF-LGN uses $w \in \{0, 0.25, 0.5\}$ (its policy weight). EASE uses $w \in \{0, 0.25, 0.5\}$, $\lambda \in \{10^3, 3 \times 10^3, 10^4\}$ and $\beta \in \{0, 0.25, 0.5\}$, which gives 27 configurations. The DAX grid has 32 configurations: prior strength $\{80, 160\}$, centre {tag, content}, lifelong strength $k \in \{0, 100\}$, tab factor on or off, and stratum cells off or on with `prior_s` $=$ 80. The gammas are $\{0, 0.05, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5, 0.7, 1.0\}$. A post-hoc arm searches the backbone trainings times 32 DAX configurations times 10 gammas, for example $12 \times 32 \times 10 = 3{,}840$ for LEAF-LGN+DAX and $27 \times 32 \times 10 = 8{,}640$ for EASE+DAX. The QP family with DAX searches 10 backbone trainings, or $10 \times 32 \times 10 = 3{,}200$, and the 1K arm searches 20, or 6,400 (Appendix Appendix A).

The tables below give the configuration chosen on validation in each split, for every method with a result file. Entries marked TBD have no result file yet. In the main run, LEAF-LGN+DAX chooses the tab factor on, the stratum cell prior 80 and gamma 0.5 in every split; the attractiveness centre is the tag mean in four splits and the content centre in split two; the lifelong strength is 100 in every split, the upper edge of its grid, and the prior strength is 80 in every split, the lower edge (Section 7). The backbone chooses learning rate 3e-3, weight decay 1e-4 and $w \in \{0.25, 0.5\}$, with 14 to 27 epochs.

Table 1: Closest works and the difference from the present work.

Work	What it estimates, and from which data	Difference from this work
SIM (Pi et al., 2020); TWIN (Chang et al., 2023)	CTR from lifelong sequences; policy logs; history and target share item ids	Exposure-conditional statistics; transfer to a disjoint pool through content keys; randomised test; SIM and TWIN not run on the main setting
Examination models (Richardson et al., 2007; Craswell et al., 2008; Chuklin et al., 2015)	Click as examination times attractiveness; position or list scan	User and scenario examination; no position term; uniform-insertion target
Dynamic Bayesian network (Chapelle and Zhang, 2009)	Perceived versus actual relevance with satisfaction	No satisfaction state
Regression-EM (Wang et al., 2018, 2016)	Position bias and relevance without randomisation; personal search	Item appeal under a uniform audience; generative fit by EM over user and tab
ExpoMF (Liang et al., 2016)	Latent exposure fitted by EM	Impressions are observed; latent examination given an impression
Unbiased LTR (Joachims et al., 2017)	Inverse-propensity weighted losses	Generative model instead of a weighted loss
Occupancy (MacKenzie et al., 2002)	Presence and detection from repeated surveys	Same likelihood structure; no recommender use found
MRP, Fay-Herriot (Park et al., 2004; Fay and Herriot, 1979)	Population and small-area estimates	Borrowed and tested on this target; weaker than DAX
NAVIP (Kim et al., 2022)	Propensity-weighted propagation in LightGCN	No lifelong data; no examination model
KuaiRand evaluations (Gao et al., 2022b; Zhang et al., 2026)	Random insertion as the test set	No out-of-pool lifelong history in the test

Table 2: Sizes of the two KuaiRand settings under the temporal protocol.

Quantity	KuaiRand-27K	KuaiRand-1K	Source
Users in both logs (dataset users)	27,077	993	run logs (27K: `dax_lgn_post.log`; 1K: `din_1k.log`)
Pool items	7,583	7,388	same
Pre-cut training clicks	523,176	18,873	`din_27k.log`; `din_1k.log`
Pre-cut training impressions	1,141,300	40,738	`v13_data_stats.txt` (CSV count)
Users with a pre-cut lifelong history	26,723	977	`din_27k.log`; `din_1k.log`
Out-of-pool impressions before T_0	135,219,168	5,016,818	`lifelong_27k_summary.txt` (27K); `v13_data_stats.txt` (1K, CSV count)
Random-exposure rows	1,186,059	42,944	`link_users.txt`; `stats_1k.txt` (993 users)
Test users (candidate lists)	27,077	993	run logs
Test users with a click, pooled	22,612	842	`stageB.txt` Section 2; `stats_1k.txt`
Test users with a click, per split	15,809 to 15,913	579 to 591	`stageB.txt` Section 2; `v13_data_stats.txt`
Candidates per user (temporal), mean (median)	43.6 (22)	43.2 (22)	`v13_data_stats.txt`

Sources are in `results/` (`v9_probes`, `v11`, `logs_v11`, `logs_v12`). The 42,944 rows are the random-log impressions of the 993 users, and the random log does not depend on the protocol. The rows marked `v13_data_stats.txt` are computed by `experiments/v13_data_stats.py` from the CSV logs and the cached test lists. The 1K out-of-pool value counts the 993 users in the CSV logs; `lifelong_1k.npz` gives 5,017,598 for the 983 users with history (the difference of 780 is not traced). The 27K value is the release summary; the npz sums to 135,219,176 (8 more).

Table 3: Leakage checks and controls, one row per unit test and per check.

Check or test	What it asserts or shows; kind and source in brackets
Unit tests in `tests/test_lifelong.py` (6 tests, all passed on 2026-11-13)	
`test_builder_uses_out_of_pool_history_strictly_before_cut`	Out-of-pool events before the cut-off only; synthetic user: 5 impressions, 2 clicks *[unit; builder]*
`test_pre_window_tables_ignore_post_cut_and_pool_rows`	Pre-cut tables unchanged when post-cut and pool rows are altered *[unit; builder]*
`test_tables_ignore_npz_rows_of_users_outside_the_dataset`	Rows of an unknown user identifier leave the tables unchanged *[unit; `leaf.py`]*
`test_lift_and_familiarity_match_the_formula`	Tables equal $(c + 5r)/(n + 5) - r$ and $\log(1 + n)$, up to scale *[unit; `leaf.py`]*
`test_labels_of_test_and_validation_never_enter_the_channels`	No label argument; flipping validation and test labels changes nDCG only *[unit; `leaf.py`]*
`test_shuffled_users_remove_the_lifelong_gain`	Synthetic gain above 0.10 falls below 0.03 in absolute value and below a quarter after shuffling users *[unit; `leaf.py`]*
Unit tests in `tests/test_dax.py` (9 tests, all passed on 2026-11-13)	
`test_em_likelihood_is_monotone_without_priors`	Observed-data likelihood does not decrease over 25 iterations (negligible priors), three seeds *[unit; `dax.py`]*
`test_em_recovers_attractiveness_and_examination_ranking`	Spearman above 0.8 for α and 0.75 for θ; tab factors within 0.08 *[unit; `dax.py`]*
`test_em_recovers_examination_up_to_the_global_scale`	Global scale removed with the true mean (test-only oracle); θ error below 0.05 *[unit; `dax.py`]*
`test_dax_fit_reads_only_pre_cut_rows_and_no_labels`	Same parameters after adding post-cut rows or flipping every random label *[unit; `dax.py`]*
`test_additive_arm_tracks_item_appeal`	Additive appeal has Spearman above 0.5 with the truth *[unit; `dax.py`]*
`test_within_tag_control_removes_the_item_gain`	Within-tag permutation keeps under 30% of the gain (stricter than gate B2) *[unit; `dax.py`]*
`test_permutations_keep_the_tag_or_not`	Within-tag permutation keeps tags; all-item permutation is a bijection *[unit; `dax.py`]*
`test_channel_interface_and_joint_module_shapes`	Channel shapes, interface and row standardisation *[unit; `dax.py`]*
`test_post_hoc_dax_reproduces_the_probe_split0`	Split-0 test value equals the probe value to 10^{-4} *[unit; `probe_pbm.json`]*
Checks in the run logs and the code	
Random log is not a training input	`training_uses_random_log = False`; fits read only the pre-cut log *[code; `data2.py`]*
Temporal boundary	Max pre-cut training time 1,650,556,797,718 ms $< T_0 \leq$ min test time 1,650,595,332,831 ms; zero leak counts asserted *[run; `dax_lgn_post.log`]*
Cached test lists equal the loader's	Test lists and users match the loader *[run; same]*
Sequence builder cut-off	Sequence events kept only if time $< T_0$ and not on a pool video (timestamp, not file name) *[code; `v12_build_seq.py`]*
Shuffled users, closed-form backbone	Full minus shuffled $+18.90$ [1.99] $\times 10^{-3}$; shuffled minus no-lifelong -1.02×10^{-3}, passes $\pm 2 \times 10^{-3}$ *[main; `stage1.txt`]*
Shuffled users, LEAF-LGN	Lifelong channels on shuffled tables: -0.93 [0.45] $\times 10^{-3}$ *[probe; `probe_27k_shuffled`]*
α permuted across all items	Gain -0.27 [0.36] $\times 10^{-3}$; removes 101.8% of the full gain (B2: at least 90%) *[main; `stageB.txt`]*
α permuted within tag	Gain $+6.28$ [1.10] $\times 10^{-3}$; a drop of 58.4% (B2: at least 30%) *[main; same]*
α tag mean; within tag; full	$+4.31$ [1.07]; $+3.29$ [0.99]; $+9.45$ [1.36] $\times 10^{-3}$ *[probe; `probe_dax_ablate`]*
α permuted across all items, weight	Selected weight 0 in 5 of 5 splits *[probe; same]*
Full ablation arm reproduces the main run	Same configurations on all five splits; same test value *[main; `dax_ablate.json`]*
Stored LEAF-LGN reproduces	Post-hoc minus stored: $+0.73$ [0.52] $\times 10^{-3}$, NB $p = 0.660$; splits agree to 5 decimals *[main; `stageB.txt`; `leaf_scores`]*
User linkage, Pure to 27K	1,186,059 random rows; identifiers identical for 0.9999 of joined events *[check; `link_users.txt`]*
Off-protocol calibration	Validation users' random rate added to LEAF-LGN: $+36.62$ [1.79] $\times 10^{-3}$; post-cut labels; not a method *[diagnostic; `probe_items.txt`]*

Table 4: Baselines, search grids and configuration counts (the n_{configs} of the run logs, as in Section 4.3). "Not run" marks a row that has no result file.

Method	Search grid	Configurations	Source
LightGCN (He et al., 2020)	lr $\{10^{-2}, 3 \cdot 10^{-3}\}$, wd $\{10^{-5}, 10^{-4}\}$; $d = 64$, 2 layers, at most 40 epochs, patience 3	4	`v10/learned_kuairand_t_base.json`
r-AdjNorm (Zhao et al., 2022)	LightGCN grid with normalisation exponent $r \in \{0.3, 0.7\}$	8	same
iALS (Hu et al., 2008) (Stage 1 only)	$\alpha \in \{3, 10, 30\}$, regularisation $\in \{1, 10, 100, 1000\}$	12	same
QP family (Appendix A)	DR-BPR-LGN-1, -p2 and -p with their own grids; per split the best of the three	20	`v10/learned_kuairand_t_qp.json`
QP family + DAX	the 10 distinct (method, configuration) trainings chosen by any split of the QP run, with the DAX grid	3,200	`v11/dax_qp.json`
EASE (Steck, 2019)	closed form, $\lambda \in \{1000, 3000, 10000\}$, $w \in \{0, 0.25, 0.5\}$, popularity $\beta \in \{0, 0.25, 0.5\}$	27	`v11/dax_ease.json`
EASE + DAX	EASE with the DAX grid	8,640	same
LEAF-LGN (post-hoc backbone)	12 trainings (lr, wd, w), no DAX	12	`v11/dax_lgn_post.json`
LEAF-LGN + DAX	12 trainings with the DAX grid and γ	3,840	same
DIN (deep interest network; Zhou et al., 2018)	4 trainings (lr, wd); dim 16, hidden 16, MLP 32, batch 1024, at most 12 epochs, patience 2; last $L = 200$ pre-cut events	4	`v12/din_27k.json`
DIN + DAX	4 trainings with the DAX grid and γ	1,280	same
SIM-hard (Pi et al., 2020)	as DIN, with the $K = 50$ most recent events sharing the tag or author of the candidate	4	`v12/sim_1k.json` (1K)
SIM-hard + DAX	as DIN + DAX	1,280	same
LEAF-tied (closed form, Stage 1 only)	λ, w, β, s and lr, wd	3,456	`v10/leaf_tied.json`
LightGCN + DAX	not run	n/a	no result file

Table 5: Methods on KuaiRand-27K (temporal, all users): mean and standard deviation of nDCG@10 over the five splits, configurations searched per split, and the source file.

Method	Mean	SD	Configurations	Source
LEAF-LGN+DAX	0.4854	0.0009	3840	dax_lgn_post.json
LEAF-LGN (post-hoc run)	0.4736	0.0002	12	dax_lgn_post.json
LEAF-LGN (stored, v10)	0.4728	0.0011	197	leaf_lgn.json
QP family	0.4634	0.0031	20	learned_kuairand_t_qp.json
QP family+DAX	0.4733	0.0012	3200	dax_qp.json
QP family (retrained)	0.4652	0.0017	10	dax_qp.json
r-AdjNorm	0.4620	0.0008	8	learned_kuairand_t_base.json
LightGCN	0.4576	0.0031	4	learned_kuairand_t_base.json
LightGCN+DAX	{TBD:results/v11/dax_lightgcn.json LightGCN+DAX mean}	{TBD:results/v11/dax_lightgcn.json LightGCN+DAX sd}	{TBD:results/v11/dax_lightgcn.json LightGCN+DAX n_configs}	{TBD:results/v11/dax_lightgcn.json LightGCN+DAX}
EASE	0.4507	0.0009	27	dax_ease.json
EASE+DAX	0.4623	0.0010	8640	dax_ease.json
DIN	0.4675	0.0007	4	din_27k.json
DIN+DAX	0.4796	0.0015	1280	din_27k.json
SIM	{TBD:results/v12/sim_27k.json SIM mean}	{TBD:results/v12/sim_27k.json SIM sd}	{TBD:results/v12/sim_27k.json SIM n_configs}	{TBD:results/v12/sim_27k.json SIM}
SIM+DAX	{TBD:results/v12/sim_27k.json SIM+DAX mean}	{TBD:results/v12/sim_27k.json SIM+DAX sd}	{TBD:results/v12/sim_27k.json SIM+DAX n_configs}	{TBD:results/v12/sim_27k.json SIM+DAX}

Configurations: post-hoc runs count backbone trainings times DAX configurations times gammas. The stored LEAF-LGN counts another unit (not comparable). QP family is the per-split best of the three stored QP variants on validation.

Table 6: Main results on KuaiRand-27K: LEAF-LGN+DAX against every comparator of the main family, with per-user and split-level (Nadeau-Bengio) tests.

Comparator	Mean	Diff. [95% hw]	p	p_{Holm}	NB p	NB p_{Holm}
LEAF-LGN+DAX (reference)	0.4854	n/a	n/a	n/a	n/a	n/a
Holm family: LEAF-LGN+DAX minus each comparator						
QP family	0.4634	+22.32 [1.83]	3.64×10^{-125}	2.18×10^{-124}	0.023	4.56×10^{-2}
QP family+DAX	0.4733	+12.04 [1.63]	3.96×10^{-47}	7.92×10^{-47}	0.006	2.23×10^{-2}
LEAF-LGN (post-hoc run)	0.4736	+11.96 [1.47]	6.55×10^{-57}	1.97×10^{-56}	0.001	6.85×10^{-3}
r-AdjNorm	0.4620	+23.79 [1.96]	2.43×10^{-123}	1.22×10^{-122}	0.000	2.73×10^{-4}
LightGCN	0.4576	+28.15 [1.86]	5.33×10^{-190}	3.73×10^{-189}	0.008	2.44×10^{-2}
DIN	0.4675	+18.26 [1.97]	1.41×10^{-73}	5.62×10^{-73}	0.001	6.62×10^{-3}
DIN+DAX	0.4796	+5.73 [1.64]	7.95×10^{-12}	7.95×10^{-12}	0.043	4.56×10^{-2}
SIM	{TBD:results/v12/sim_27k.json SIM mean}	{TBD:results/v12/sim_27k.json SIM paired difference}	{TBD:results/v12/sim_27k.json SIM}	{TBD:results/v12/sim_27k.json SIM}	{TBD:results/v12/sim_27k.json SIM}	{TBD:results/v12/sim_27k.json SIM}
SIM+DAX	{TBD:results/v12/sim_27k.json SIM+DAX mean}	{TBD:results/v12/sim_27k.json SIM+DAX paired difference}	{TBD:results/v12/sim_27k.json SIM+DAX}	{TBD:results/v12/sim_27k.json SIM+DAX}	{TBD:results/v12/sim_27k.json SIM+DAX}	{TBD:results/v12/sim_27k.json SIM+DAX}
Outside the family (raw p values only; not Holm-adjusted)						
EASE	0.4507	+35.24 [2.24]	6.53×10^{-205}	n/a	0.000	n/a
EASE+DAX	0.4623	+23.26 [1.98]	1.52×10^{-115}	n/a	0.001	n/a

Per-user columns: paired t test over the pooled test users (n = 22612); Holm within the family, over the 7 comparators present; NB: Nadeau-Bengio corrected resampled t test over the five split means (fraction of validation users 0.3); Holm over the same family. Missing family members (shown as TBD): SIM, SIM+DAX. Differences are per-user means x1e-3 nDCG@10 with the 95% half-width.

Table 7: Sensitivity of the Holm adjustment: the main family extended by EASE and EASE+DAX (the family size is given in the column header and counts the comparators whose files exist).

Comparator	p	p_{Holm} (family of 9)	NB p	NB p_{Holm}
QP family	3.64×10^{-125}	2.55×10^{-124}	0.023	4.56×10^{-2}
QP family+DAX	3.96×10^{-47}	7.92×10^{-47}	0.006	2.23×10^{-2}
LEAF-LGN (post-hoc run)	6.55×10^{-57}	1.97×10^{-56}	0.001	6.85×10^{-3}
r-AdjNorm	2.43×10^{-123}	1.46×10^{-122}	0.000	3.52×10^{-4}
LightGCN	5.33×10^{-190}	4.27×10^{-189}	0.008	2.44×10^{-2}
DIN	1.41×10^{-73}	5.62×10^{-73}	0.001	6.62×10^{-3}
DIN+DAX	7.95×10^{-12}	7.95×10^{-12}	0.043	4.56×10^{-2}
SIM	{TBD:results/v12/sim_27k.json SIM paired test}	{TBD:results/v12/sim_27k.json SIM paired test}	{TBD:results/v12/sim_27k.json SIM paired test}	{TBD:results/v12/sim_27k.json SIM paired test}
SIM+DAX	{TBD:results/v12/sim_27k.json SIM+DAX paired test}	{TBD:results/v12/sim_27k.json SIM+DAX paired test}	{TBD:results/v12/sim_27k.json SIM+DAX paired test}	{TBD:results/v12/sim_27k.json SIM+DAX paired test}
EASE	6.53×10^{-205}	5.88×10^{-204}	0.000	1.16×10^{-3}
EASE+DAX	1.52×10^{-115}	7.61×10^{-115}	0.001	5.39×10^{-3}

The Holm values differ from Table 6 only through the family size.

Table 8: Fair-baseline reading: the DAX increments and the stored LEAF-LGN against the QP family, with their search budgets (configurations of the first and second method).

Comparison	Configurations	Diff. [95% hw]	p	NB p	Test family
LEAF-LGN+DAX minus QP family+DAX (fair baseline)	3840 / 3200	+12.04 [1.63]	3.96×10^{-47}	0.006	main family
QP family+DAX minus QP family (DAX on the QP backbone)	3200 / 20	+10.28 [1.37]	1.09×10^{-48}	0.105	raw
LEAF-LGN+DAX minus LEAF-LGN (DAX on the LEAF backbone)	3840 / 12	+11.96 [1.47]	6.55×10^{-57}	0.001	main family
LEAF-LGN (stored, v10) minus QP family	197 / 20	+9.63 [1.60]	6.77×10^{-32}	0.063	raw

Rows marked raw are not Holm-adjusted: the stored v10 comparison is part of a different family (eight comparisons in the stage 1 run), and the DAX increments are reported beside the main family. The QP family + DAX search uses only the QP backbone configurations selected without DAX.

Table 9: DAX is backbone-agnostic: nDCG@10 without and with the post-hoc DAX module on KuaiRand-27K, paired difference x1e-3 with the 95% half-width, two-sided t test and the Nadeau-Bengio test over the split means.

Backbone	Without DAX	With DAX	Diff. [95% hw]	p	NB p
EASE	0.4507	0.4623	+11.98 [1.64]	2.27×10^{-46}	0.000
LightGCN	0.4576	{TBD:results/v11/dax_lightgcn.json LightGCN+DAX mean}	{TBD:results/v11/dax_lightgcn.json LightGCN+DAX minus results/v10/learned_kuairand_1_base.json LightGCN paired difference}		
QP family (stored)	0.4634	0.4733	+10.28 [1.37]	1.09×10^{-48}	0.105
LEAF-LGN (post-hoc run)	0.4736	0.4854	+11.96 [1.47]	6.55×10^{-57}	0.001
DIN	0.4675	0.4796	+12.53 [1.41]	1.30×10^{-67}	0.011
SIM	{TBD:results/v12/sim_27k.json SIM mean}	{TBD:results/v12/sim_27k.json SIM+DAX mean}	{TBD:results/v12/sim_27k.json SIM+DAX minus SIM paired difference}		

Raw p values (no family). The QP row uses the stored QP family as the reference, as in the fair-baseline reading.

Table 10: DAX ablations on KuaiRand-27K (post-hoc): gain over the backbone of the ablation run, the difference to full DAX, and the shuffle controls.

Arm	Configs	Mean	Gain [95% hw]	p_{Holm}	Kept (%)	Full minus arm	p_{Holm}
full DAX	320	0.4854	+15.10 [1.42]	8.82×10^{-95}	100.0	n/a	n/a
no examination factor kappa_t	160	0.4839	+13.59 [1.43]	1.28×10^{-76}	90.0	+1.51 [0.79]	2.02×10^{-4}
no lifelong prior on theta_u	160	0.4836	+13.31 [1.43]	7.92×10^{-73}	88.2	+1.78 [0.61]	2.16×10^{-8}
no tag prior on alpha_i	160	0.4820	+11.68 [1.22]	1.28×10^{-76}	77.4	+3.41 [0.99]	4.79×10^{-11}
additive logit (MRP-style), tab effects	320	0.4794	+9.01 [1.25]	9.20×10^{-45}	59.7	+6.09 [0.96]	5.43×10^{-35}
alpha shuffled across items	960	0.4704	-0.27 [0.36]	1.48×10^{-1}	-1.8	+15.36 [1.40]	3.53×10^{-101}
alpha shuffled within tag	960	0.4768	+6.28 [1.10]	1.23×10^{-28}	41.6	+8.81 [1.00]	2.47×10^{-65}
Permutation seeds (pooled arm above selects over seeds; separate runs below)							
alpha shuffled across items, seed 0			-0.44 [0.50]	NB 0.510	-2.9		
alpha shuffled across items, seed 1			-0.16 [0.53]	NB 0.986	-1.0		
alpha shuffled across items, seed 2			-0.20 [0.51]	NB 0.775	-1.3		
alpha shuffled within tag, seed 0			+5.66 [1.14]	NB 0.010	37.5		
alpha shuffled within tag, seed 1			+6.84 [1.13]	NB 0.002	45.3		
alpha shuffled within tag, seed 2			+6.35 [1.24]	NB 0.001	42.1		

Gain: arm minus the LEAF-LGN base of the ablation run (mean 0.4706, 2 backbone trainings, five gammas, epoch window of 3 around the main selection). Full DAX reproduces the main run's splits. Holm within the seven gains (column p Holm); the column of full minus arm is Holm-adjusted within the six ablated arms. Kept: the arm's gain as a percentage of the full gain. Pooled shuffle entries select the permutation seed on validation.

Table 11: KuaiRand-1K replication: LEAF-LGN+DAX against the QP family with and without DAX (nDCG@10).

Method or comparison	Mean	SD	Configs	Diff. [95% hw]	p	p_{Holm}	NB p	NB p_{Holm}
LEAF-LGN+DAX	0.4590	0.0038	3840					
QP family	0.4246	0.0043	20					
QP family+DAX	0.4358	0.0035	6400					
LEAF-LGN+DAX minus QP family				+37.58 [11.10]	5.37×10^{-11}	1.61×10^{-10}	0.011	2.28×10^{-2}
LEAF-LGN+DAX minus QP family+DAX				+25.82 [9.66]	1.96×10^{-7}	3.91×10^{-7}	0.007	2.18×10^{-2}
QP family+DAX minus QP family				+11.76 [8.54]	7.04×10^{-3}	7.04×10^{-3}	0.160	1.60×10^{-1}

KuaiRand-1K: 842 test users pooled over five splits (each split 30% validation, 70% test); 1K users are a subset of the 27K users, a lower-power replication. The 1K QP family + DAX arm searches all 20 QP configurations (6400 configurations in total) whereas the 27K arm searches the 10 selected ones (3200); this favours the baseline and is stated as such. Holm over the three comparisons of this table; NB over the split means. Split differences (x1e-3): LEAF-LGN+DAX minus QP family: +36.7 +25.7 +35.6 +37.2 +36.6; LEAF-LGN+DAX minus QP family+DAX: +21.3 +20.3 +23.7 +23.1 +27.8; QP family+DAX minus QP family: +15.4 +5.4 +11.9 +14.1 +8.8.

Table 12: Training seeds on KuaiRand-27K: LEAF-LGN+DAX and its within-seed gain over LEAF-LGN (same seed), seeds 0 to 3.

Seed	LEAF-LGN+DAX	LEAF-LGN	Configs	Gain [95% hw]	p	NB p
0 (main run)	0.4854	0.4736	3840	+11.96 [1.47]	6.55×10^{-57}	0.001
1	0.4858	0.4714	640	+14.67 [1.42]	1.63×10^{-90}	0.001
2	0.4856	0.4708	640	+15.10 [1.46]	1.87×10^{-90}	0.002
3	0.4844	0.4716	640	+12.89 [1.44]	1.90×10^{-68}	0.002
Spread over seeds	0.4853 ± 0.0006			+13.65 ± 1.48		
QP family (seeds not run)	{TBD:results/v11/c_seeds QP seeds 1-3: qp_s1.json.partial.json incomplete}					

Seed 0 is the main run with the full backbone grid (12 trainings). Seeds 1 to 3 re-train with the training seed changed and a reduced backbone grid: lr 3e-3 and wd 1e-4 in every split, w in 0.25, 0.5 for the DAX arm (2 trainings) and w 0.25 for the base. The epoch, the DAX configuration and gamma are selected on validation per split. Raw p values, not Holm-adjusted. The QP family was not re-trained under seeds 1 to 3, so no seed comparison with it is reported.

Table 13: Lifelong deep baselines (DIN and SIM-hard) with and without DAX, and their paired difference to LEAF-LGN+DAX (LEAF-LGN+DAX minus the comparator, x1e-3 nDCG@10).

Comparator	Mean	Configs	Diff. [95% hw]	p	NB p
KuaiRand-27K					
DIN	0.4675	4	+18.26 [1.97]	1.41×10^{-73}	0.001
DIN+DAX	0.4796	1280	+5.73 [1.64]	7.95×10^{-12}	0.043
SIM	{TBD:results/v12/sim_27k.json SIM mean}	{TBD:results/v12/sim_27k.json SIM configs}	{TBD:results/v12/sim_27k.json SIM difference}		
SIM+DAX	{TBD:results/v12/sim_27k.json SIM+DAX mean}	{TBD:results/v12/sim_27k.json SIM+DAX configs}	{TBD:results/v12/sim_27k.json SIM+DAX difference}		
KuaiRand-1K					
DIN	0.4189	4	+39.26 [10.37]	2.63×10^{-13}	0.002
DIN+DAX	0.4228	1280	+35.89 [9.25]	7.10×10^{-14}	0.002
SIM	0.4246	4	+35.26 [9.24]	1.70×10^{-13}	0.007
SIM+DAX	0.4357	1280	+23.29 [8.69]	1.81×10^{-7}	0.095

Lifelong sequence models: DIN attends over the last 200 lifelong events; SIM-hard keeps the 50 most recent events sharing the candidate's tag or author.

Raw p values. DIN and SIM are tuned over four configurations (lr 1e-2, 3e-3 times wd 1e-4, 1e-5) with patience 2; their +DAX arms use the post-hoc DAX grid. Entries marked TBD have no result file yet.

Table 14: LEAF controls on KuaiRand-27K (stage 1 stored run, LEAF-tied backbone): arm means and full minus arm.

Arm	Configs	Mean	Full minus arm [95% hw]	p_{Holm}	NB p	NB p_{Holm}
LEAF-tied (full)	3456	0.4691	n/a	n/a	n/a	n/a
shuffle users	3456	0.4499	+18.90 [1.99]	4.90×10^{-76}	0.001	4.60×10^{-3}
no tag channels	1152	0.4594	+9.38 [1.63]	1.17×10^{-28}	0.002	6.48×10^{-3}
no author channels	432	0.4628	+6.45 [1.17]	2.10×10^{-26}	0.001	4.42×10^{-3}
no lifelong channels	72	0.4512	+17.88 [2.04]	9.61×10^{-65}	0.000	2.51×10^{-3}
no policy channel	864	0.4643	+5.01 [1.20]	6.91×10^{-16}	0.066	1.32×10^{-1}
lift only	432	0.4634	+5.78 [1.11]	6.10×10^{-24}	0.001	4.60×10^{-3}
familiarity only	576	0.4585	+10.26 [1.66]	4.07×10^{-33}	0.001	4.60×10^{-3}
no music channels	1728	0.4691	0.00 [0.00]	1	1.000	1

Full minus arm: positive means the channel helps. Holm within the eight arms; NB over the split means. The no-music arm gives identical rankings in every split. The LEAF-LGN ablations are not run (results/v10/leaf_lgn_ablate.json is missing); these controls use the closed-form LEAF-tied backbone. The shuffle-users control removes the lifelong gain (design check on the difference between the shuffled and the no-lifelong arm).

Table C.15: Chosen configurations per split: LEAF-LGN+DAX (main, post-hoc) (nDCG@10 mean 0.4854).

Split	Chosen configuration
1	`lr=0.003, wd=0.0001, w=0.25, epoch=23, gamma=0.5, prior=80, centre=tag, k_life=100, tab=True, prior_s=80`
2	`lr=0.003, wd=0.0001, w=0.5, epoch=25, gamma=0.5, prior=80, centre=content, k_life=100, tab=True, prior_s=80`
3	`lr=0.003, wd=0.0001, w=0.5, epoch=27, gamma=0.5, prior=80, centre=tag, k_life=100, tab=True, prior_s=80`
4	`lr=0.003, wd=0.0001, w=0.25, epoch=14, gamma=0.5, prior=80, centre=tag, k_life=100, tab=True, prior_s=80`
5	`lr=0.003, wd=0.0001, w=0.5, epoch=27, gamma=0.5, prior=80, centre=tag, k_life=100, tab=True, prior_s=80`

Table C.16: Chosen configurations per split: LEAF-LGN (post-hoc base) (nDCG@10 mean 0.4736).

Split	Chosen configuration
1	`lr=0.003, wd=0.0001, w=0, epoch=25`
2	`lr=0.003, wd=0.0001, w=0, epoch=18`
3	`lr=0.003, wd=0.0001, w=0, epoch=38`
4	`lr=0.003, wd=1e-05, w=0, epoch=21`
5	`lr=0.003, wd=0.0001, w=0, epoch=17`

Table C.17: Chosen configurations per split: QP family (stored, per-split best variant) (nDCG@10 mean 0.4634).

Split	Chosen configuration
1	lr=0.003, wd=0.0001, dim=64, layers=2, fn=1, epoch=38
2	lr=0.01, wd=0.0001, dim=64, layers=2, fn=1, epoch=12
3	lr=0.01, wd=0.0001, dim=64, layers=2, fn=1, epoch=11
4	lr=0.003, wd=0.0001, dim=64, layers=2, fn=1, epoch=39
5	lr=0.01, wd=1e-05, dim=64, layers=2, fn=1, epoch=3

Table C.18: Chosen configurations per split: QP family+DAX (nDCG@10 mean 0.4733).

Split	Chosen configuration
1	method=DR-BPR-LGN-1, lr=0.003, wd=0.0001, dim=64, layers=2, fn=1, epoch=39, gamma=0.5, prior=80, centre=tag, k_life=100, tab=True, prior_s=None
2	method=DR-BPR-LGN-1, lr=0.01, wd=0.0001, dim=64, layers=2, fn=1, epoch=12, gamma=0.5, prior=80, centre=tag, k_life=100, tab=True, prior_s=None
3	method=DR-BPR-LGN-1, lr=0.003, wd=0.0001, dim=64, layers=2, fn=1, epoch=39, gamma=0.5, prior=80, centre=tag, k_life=100, tab=True, prior_s=None
4	method=DR-BPR-LGN-1, lr=0.01, wd=0.0001, dim=64, layers=2, fn=1, epoch=19, gamma=0.5, prior=80, centre=tag, k_life=100, tab=True, prior_s=None
5	method=DR-BPR-LGN-1, lr=0.01, wd=1e-05, dim=64, layers=2, fn=1, epoch=14, gamma=0.7, prior=80, centre=tag, k_life=100, tab=True, prior_s=None

Table C.19: Chosen configurations per split: EASE (nDCG@10 mean 0.4507).

Split	Chosen configuration
1	w=0.5, epoch=0, lam=10000.0, beta=0.0
2	w=0.5, epoch=0, lam=10000.0, beta=0.0
3	w=0.5, epoch=0, lam=10000.0, beta=0.0
4	w=0.5, epoch=0, lam=10000.0, beta=0.0
5	w=0.5, epoch=0, lam=10000.0, beta=0.0

Table C.20: Chosen configurations per split: EASE+DAX (nDCG@10 mean 0.4623).

Split	Chosen configuration
1	w=0.5, epoch=0, gamma=0.15, prior=80, centre=tag, k_life=100, tab=True, prior_s=None, lam=10000.0, beta=0.0
2	w=0.5, epoch=0, gamma=0.15, prior=80, centre=tag, k_life=100, tab=True, prior_s=80, lam=10000.0, beta=0.0
3	w=0.5, epoch=0, gamma=0.2, prior=80, centre=tag, k_life=100, tab=True, prior_s=None, lam=10000.0, beta=0.0
4	w=0.5, epoch=0, gamma=0.15, prior=80, centre=tag, k_life=100, tab=True, prior_s=None, lam=10000.0, beta=0.0
5	w=0.5, epoch=0, gamma=0.15, prior=80, centre=tag, k_life=100, tab=True, prior_s=None, lam=10000.0, beta=0.0

Table C.21: Chosen configurations per split: LightGCN (nDCG@10 mean 0.4576).

Split	Chosen configuration
1	lr=0.003, wd=0.0001, dim=64, layers=2, epoch=38
2	lr=0.01, wd=0.0001, dim=64, layers=2, epoch=14
3	lr=0.01, wd=0.0001, dim=64, layers=2, epoch=6
4	lr=0.01, wd=0.0001, dim=64, layers=2, epoch=12
5	lr=0.01, wd=1e-05, dim=64, layers=2, epoch=3

Table C.22: Chosen configurations per split: r-AdjNorm (nDCG@10 mean 0.4620).

Split	Chosen configuration
1	lr=0.01, wd=0.0001, dim=64, layers=2, r=0.7, epoch=17
2	lr=0.01, wd=0.0001, dim=64, layers=2, r=0.7, epoch=15
3	lr=0.01, wd=0.0001, dim=64, layers=2, r=0.7, epoch=11
4	lr=0.01, wd=1e-05, dim=64, layers=2, r=0.7, epoch=18
5	lr=0.01, wd=0.0001, dim=64, layers=2, r=0.7, epoch=12

Table C.23: Chosen configurations per split: DIN (27K) (nDCG@10 mean 0.4675).

Split	Chosen configuration
1	lr=0.01, wd=1e-05, epoch=9
2	lr=0.01, wd=1e-05, epoch=9
3	lr=0.01, wd=0.0001, epoch=7
4	lr=0.01, wd=0.0001, epoch=7
5	lr=0.01, wd=1e-05, epoch=8

Table C.24: Chosen configurations per split: DIN+DAX (27K) (nDCG@10 mean 0.4796).

Split	Chosen configuration
1	`lr=0.01, wd=1e-05, epoch=9, gamma=0.4, prior=80, centre=tag, k_life=100, tab=True, prior_s=None`
2	`lr=0.01, wd=1e-05, epoch=9, gamma=0.4, prior=80, centre=tag, k_life=100, tab=True, prior_s=80`
3	`lr=0.003, wd=0.0001, epoch=10, gamma=0.4, prior=80, centre=tag, k_life=100, tab=True, prior_s=None`
4	`lr=0.01, wd=0.0001, epoch=7, gamma=0.4, prior=80, centre=tag, k_life=100, tab=True, prior_s=None`
5	`lr=0.01, wd=1e-05, epoch=9, gamma=0.4, prior=80, centre=tag, k_life=100, tab=True, prior_s=80`

Table C.25: Chosen configurations, SIM (27K): not available.

Split
{TBD:SIM (27K) chosen configurations}

Table C.26: Chosen configurations, SIM+DAX (27K): not available.

Split
{TBD:SIM+DAX (27K) chosen configurations}

Table C.27: Chosen configurations per split: LEAF-LGN+DAX (1K) (nDCG@10 mean 0.4590).

Split	Chosen configuration
1	`lr=0.01, wd=0.0001, w=0.25, epoch=0, gamma=0.3, prior=160, centre=tag, k_life=0, tab=False, prior_s=80`
2	`lr=0.01, wd=0.0001, w=0.5, epoch=0, gamma=0.5, prior=80, centre=tag, k_life=100, tab=False, prior_s=None`
3	`lr=0.01, wd=0.0001, w=0.5, epoch=0, gamma=0.5, prior=160, centre=tag, k_life=100, tab=False, prior_s=80`
4	`lr=0.01, wd=0.0001, w=0.5, epoch=3, gamma=0.2, prior=80, centre=content, k_life=100, tab=True, prior_s=None`
5	`lr=0.01, wd=0.0001, w=0.5, epoch=1, gamma=0.15, prior=80, centre=content, k_life=100, tab=False, prior_s=80`

Table C.28: Chosen configurations per split: QP family+DAX (1K) (nDCG@10 mean 0.4358).

Split	Chosen configuration
1	method=DR-BPR-LGN-1, lr=0.01, wd=0.0001, dim=64, layers=2, fn=1, epoch=4, gamma=1, prior=160, centre=tag, k_life=100, tab=True, prior_s=80
2	method=DR-BPR-LGN-p, lr=0.01, wd=0.0001, dim=64, layers=2, fn=1, pbeta=0.5, epoch=37, gamma=1, prior=160, centre=tag, k_life=100, tab=True, prior_s=80
3	method=DR-BPR-LGN-p, lr=0.01, wd=1e-05, dim=64, layers=2, fn=1, pbeta=0.5, epoch=31, gamma=0.4, prior=80, centre=tag, k_life=100, tab=False, prior_s=80
4	method=DR-BPR-LGN-p, lr=0.01, wd=0.0001, dim=64, layers=2, fn=1, pbeta=0.5, epoch=33, gamma=0.7, prior=80, centre=tag, k_life=100, tab=False, prior_s=80
5	method=DR-BPR-LGN-1, lr=0.01, wd=0.0001, dim=64, layers=2, fn=1, epoch=10, gamma=0.7, prior=80, centre=tag, k_life=100, tab=True, prior_s=None

Table C.29: Chosen configurations per split: DIN (1K) (nDCG@10 mean 0.4189).

Split	Chosen configuration
1	lr=0.003, wd=0.0001, epoch=0
2	lr=0.003, wd=0.0001, epoch=0
3	lr=0.003, wd=0.0001, epoch=0
4	lr=0.01, wd=1e-05, epoch=1
5	lr=0.003, wd=0.0001, epoch=0

Table C.30: Chosen configurations per split: DIN+DAX (1K) (nDCG@10 mean 0.4228).

Split	Chosen configuration
1	lr=0.003, wd=0.0001, epoch=0, gamma=0.7, prior=160, centre=tag, k_life=100, tab=False, prior_s=None
2	lr=0.01, wd=0.0001, epoch=1, gamma=1, prior=160, centre=tag, k_life=100, tab=True, prior_s=80
3	lr=0.003, wd=0.0001, epoch=0, gamma=0.7, prior=160, centre=tag, k_life=100, tab=False, prior_s=None
4	lr=0.01, wd=1e-05, epoch=1, gamma=0.4, prior=80, centre=tag, k_life=100, tab=False, prior_s=80
5	lr=0.003, wd=0.0001, epoch=1, gamma=1, prior=160, centre=tag, k_life=100, tab=True, prior_s=None

Table C.31: Chosen configurations per split: SIM (1K) (nDCG@10 mean 0.4246).

Split	Chosen configuration
1	`lr=0.003, wd=0.0001, epoch=0`
2	`lr=0.01, wd=0.0001, epoch=1`
3	`lr=0.003, wd=0.0001, epoch=0`
4	`lr=0.003, wd=1e-05, epoch=7`
5	`lr=0.003, wd=0.0001, epoch=1`

Table C.32: Chosen configurations per split: SIM+DAX (1K) (nDCG@10 mean 0.4357).

Split	Chosen configuration
1	`lr=0.003, wd=0.0001, epoch=0, gamma=0.2, prior=160, centre=content, k_life=100, tab=False, prior_s=None`
2	`lr=0.01, wd=1e-05, epoch=1, gamma=0.4, prior=160, centre=tag, k_life=0, tab=True, prior_s=80`
3	`lr=0.01, wd=1e-05, epoch=1, gamma=0.3, prior=160, centre=tag, k_life=100, tab=True, prior_s=None`
4	`lr=0.003, wd=0.0001, epoch=6, gamma=0.5, prior=80, centre=tag, k_life=0, tab=False, prior_s=None`
5	`lr=0.01, wd=1e-05, epoch=1, gamma=0.5, prior=160, centre=tag, k_life=100, tab=True, prior_s=80`